Class 9

Physics

Volume 2

Solutions

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4. Fluid Mechanics

WORKSHEET – 1

Level – I

1) C	2) C	3) D	4) B	5) A	6) B	7) C	8) A
9) B	10) C						

Level – II

1) A	2) C	3) B	4) B	5) B	6) A	7) B	8) A
9) D	10) A	11) D	12) D	13) A	14) A	15) A	16) C
17) A	18) B	19) a-s; b-p; c-r; d-q		20) C	21) A,D	22) A,B	
23) D	24) C	25) B					

1. [Option - A]
2. $1\ g/cm^3 = 1000\ kg/m^3$
   In. C.G.S. R.D = 0.8
   In. S.I R.D = $800\ kg/m^3$
   [option - C]
3. $\text{volume} = \frac{\text{mass}}{\text{density}} = \frac{900}{3} = 300\ cm^3$
   [option - B]
4. since volume is same
   $$\rho = \frac{\rho_1 + \rho_2}{2} = \frac{1 + 0.85}{2}$$
   $= 0.925\ g/cm^3 = 0.925 \times 10^3\ kg/m^3$
   [option - B]
5. $p = \frac{F}{A} = \frac{10}{20} = 0.5\ N/m^2$
   [option - B]
6. $p = \frac{F}{A} = \frac{50}{10} = 5 dyne/cm^2$
   [option - A]
7. $\text{volume} = \frac{\text{mass}}{\text{density}} = \frac{800}{4} = 200\ cm^3$
   [option - B]
8. [option - A]
9. $\rho_{mix} = \frac{2\rho_1\rho_2}{\rho_1+\rho_2}$
   $\rho_{mix} = 1.52 \times 10^3$
   [option - D]
10. $\rho_{mix} = \frac{2(\rho)(2\rho)}{\rho+2\rho} = \frac{4\rho}{3}$ [option - A]
11. $\rho_{mix} = \frac{\rho_1+\rho_2}{2} = \frac{\rho+2\rho}{2} = \frac{3\rho}{2}$ [option - D]
12. [option - D]
13. $p = \frac{F}{A}$; $F = P \cdot A$
    $= 10 \times 25 \times 10^{-4} = 2.5 \times 10^{-4}\ N$ [option - A]
14. $F = P.A = 10 \times 100 = 1000N$
    [option - 1]
15. $\rho_{mix} = \frac{2\rho_1\rho_2}{\rho_1+\rho_2} = \frac{2\times3\times4}{7} = \frac{24}{7}$ [option - A]
16. $\rho_{mix} = \frac{\text{Total mass}}{\text{Total volume}}$

$= \frac{2(5\times10^{-3})+3(6\times10^{-3})}{(5+6)\times10^{-3}} = \frac{10+18}{11} = \frac{28}{11}$

[option - C]

17. $m_1 = 2m, m_2 = 3m$

$\rho_{\text{mix}} = \frac{\text{Total mass}}{\text{Total volume}}$

$= \frac{5m}{\frac{2m}{3}+\frac{3m}{8}} = \frac{5m}{\left(\frac{25}{24}\right)}$

= 4.8 g/cc [option - A]

18. $v_1 = v, v_2 = 2v$

$\rho_{\text{mix}} = \frac{\text{Total mass}}{\text{Total volume}}$

$= \frac{2(v)+3(2v)}{v+2v} = \frac{2(v)+6v}{3v} = \frac{8v}{3v} = \frac{8}{3}$ [option - B]

19. $a-s, b-p, c-v, d-q.$

20. [option - C]

21. if masses are equal $\rho_{\text{mix}} = \frac{2(\rho)\left(\frac{\rho}{2}\right)}{\rho+\frac{\rho}{2}} = \frac{\rho^2}{3\left(\frac{\rho}{2}\right)} = \frac{2\rho}{3}$

if volumes are equal

$$\rho_{\text{mix}} = \frac{\rho+\frac{\rho}{2}}{2} = \frac{3\rho}{4}$$ [option - A,D]

22. [option - A,B]

23. $\rho_{\text{mix}} = \frac{2(2)(6)}{2+6} = \frac{24}{8} = 3$ kg/m^3 [option - D]

24. $\rho_{\text{mix}} = \frac{2+6}{2} = 4$ kg/m^3 [option - C]

25. $\rho_{\text{mix}} = \frac{\text{Total mass}}{\text{Total volume}} = \frac{m+2m}{\frac{m}{2}+\frac{2m}{6}} = \frac{3m\times12}{10m} = \frac{18}{5}$ [option - B]

WORKSHEET – 2

Level – I

1) C	2) A	3) B	4) B	5) D	6) A	7) C
8) A	9) D	10) C				

Level – II

1) A	2) B	3) C	4) B	5) A	6) A	7) A
8) D	9) B	10) a-r; b-p,q; c-s; d-t		11) A	12) B	13) B
14) B	15) A	16) A,B	17) a-p; b-q; c-r; d-q		18) a-p; b-s; c-q; d-r	
19) A	20) B	21) C	22) 1-b; 2-a; 3-d; 4-c		23) 41	24) B
25) C						

1. $F_1 = 900\text{N} d_1 = 1.8$ cm $\Rightarrow r_1 = 0.9$ cm $= 0.9\times10^{-2}$ m

$F_2 = ?\, d_2 = 36$ cm $\Rightarrow r_2 = 18$ cm $= 18\times10^{-2}$ m

$\frac{F_1}{A_1} = \frac{F_1}{A_2} \Rightarrow F_2 = \frac{F_1}{A_1}\times A_2 = F_1 \times \frac{r_2^2}{r_1^2}$

$= \frac{900\times(18\times10^{-2})^2}{(0.9\times10^{-2})^2} = 36\times10^4$ N

2. $A_1 = 4$ m^2 $F_1 = ?$

$A_2 = 8m^2$ $F_2 = 800$kgf

$\frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow F_1 = F_2 \times \frac{A_1}{A_2} = 800\times\frac{4}{8} = 400$kgf

3. $P_B - P_A = \rho g\ h = 10^3 \times 10 \times 10 \times 10^{-2}$
$P_B - P_A = 1000\ N/m^2$

4. $\frac{r_1}{r_2} = \frac{50}{4}\ F_1 = ?\ F_2 = 20kgf$

$$F_1 = \left(\frac{A_1}{A_2}\right)F_2 = \left(\frac{r_1^2}{r_2^2}\right)F_2$$

$$F_1 = \left(\frac{50}{4}\right)^2 \times 20 = \frac{625}{4} \times 20$$

$$F_1 = 3125kgf$$

5. $P = P_o + \rho gh$

$$= P_0 + \frac{m}{v} \times g \times h = P_0 + \frac{m}{I \times b \times h} \times g \times h$$

$$P = P_0 + \frac{mg}{A}$$

6. $h = 20\ cm r = 10\ cm,\ F = ?$

$$P = P_0 + \rho gh$$
$$= (1.01 \times 10^5) + [1000 \times 10 \times 20 \times 10^{-2}]$$
$$= (10^5 \times 1.01) + [0.02 \times 10^5]$$
$$= 10^5[1.01 + 0.02] = 1.03 \times 10^5\ Pa$$
$$p = \frac{F}{A} \Rightarrow F = PA = P\pi r^2$$
$$= 1.0.3 \times 10^5 \times \frac{22}{7} \times (10 \times 10^{-2})^2$$
$$= 1.03 \times 10^5 \times 3.14 \times 100 \times 10^{-4} = 3230\ N$$

7. $P = \rho_1 gh_1 = 4 \times 9.8 \times 10^3 = 3.92 \times 10^4\ Pa$
$P_2 = \rho_2\ h_2\ g = 1.7 \times 10^3 \times 4 \times 9.8 = 6.664 \times 10^4\ Pa$
$\Delta P = P_2 - P_1$ (Pressure diffrence b/w water and acid columns)

$$\Delta P = (6.664 - 3.92)10^4 = 2.744 \times 10^4\ Pa$$

$\therefore F = \Delta P.A$
$F = 2.744 \times 10^4 \times 20 \times 10^{-4} = 54.88\ N$

8. Pressure on Smaller Piston $= \frac{F}{A} = \frac{mg}{\pi r^2}$

$$= \frac{20 \times 9.8}{\pi \times (5 \times 10^{-2})^2}\ N \setminus m^2$$

Pressure on larger piston $= \frac{F}{\pi \times r^2} = \frac{F}{\pi(17.5 \times 10^{-2})} N \setminus m^2$
The difference between the two Pressures = heg
$h = 1.5\ m \& P = 750Kg \setminus m^3$

$$P_2 - P_1 = Pgh$$
$$\frac{20 \times 9.8}{\pi \times 25 \times 10^{-4}} - \frac{F}{\pi \times 17.5 \times 17.5 \times 10^{-4}} = 1.5 \times 750 \times 9.8$$

by solving $F = 1.3 \times 10^3\ N$

9. $\frac{W + \rho ghs}{s} = \frac{F}{\frac{s}{10}};\ F = \frac{W + \rho ghs}{10}$

10. $a \to r; b \to p, q; c \to s; d \to t$

11. $F = 30\ N\ S = 10\ m\ \ W = ?$
$W = FS = 30 \times 10 = 300\ J$

12. $A_1 = 5\ cm^2\ F_1 = ?$

$$A_2 = 625\ cm^2 F_2 = 1250\ N$$
$$\frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow F_1\left(\frac{A_1}{A_2}\right)F_2 = \left(\frac{5}{625}\right) \times 1250 = 10N$$

13. $A_1 = 2\ cm^2\ \ F_1 = 10\ N$

$A_2 = 20\ cm^2\ F_2 = ?$

$$F_2 = \left(\frac{A_2}{A_1}\right)F_1 = \left(\frac{20}{2}\right)10 = 100\ N$$

14. $A_1 = 1\ cm^2\ \ F_1 = 5\ N$

$A_2 = 10\ cm^2\ F_2 = ?$

$$\frac{F_1}{A_1} = \frac{F_2}{A_2} > F_2 = \left(\frac{A_2}{A_1}\right)F_1$$

$$F_2 = \frac{10}{1} \times 5 = 50\ N$$

$F_2 = 50\ N$ (or) 50×10^5 dyne

15. (OPTION -A)

16. (OPTION −A&B)

17. $A_1 = 5 \times 10^{-4}\ m^2;\ A_2 = 10^{-2}\ m^2$

$F_1 = 20\ N$

a) $P = \frac{F}{A} = \frac{20}{5\times10^{-4}} = 4 \times 10^4 = 40000 P_a$

b) F2 = ?

$$\frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow F_2 = F_1 \times \frac{A_2}{A_1}$$

$$F_2 = 20 \times \frac{10^{-2}}{5 \times 10^{-4}} = 4 \times 10^2 = 400\ N$$

c) $w = Fs$

$= 20 \times 2 \times 10^{-4} = 40 \times 10^{-4}$

$= 40 \times 10^{-3}\ J$

$= 4 \times 10^{-3} \times 10^7 = 4 \times 10^7 erg$

$W = FS; = 20 \times 20 = 400\ J$

18. $a - p; b - s; c - q; d - r$

19. $p = p_0 + \rho gh$

$p = 1\ atm + 10^3 \times 1.03 \times 10^3 \times 10$

$p = 1\ atm + 103\ atm\ ; p = 104\ atm$

20. Guagepressure $= p - p_0$

Guagepressure $= 104\ atm - 1\ atm$

Guagepressure $= 103\ atm$

21. $F = P_{guage} \times A;\ F = 103 \times 10^5 \times 400 \times 10^{-4}$

$F = 4.12 \times 10^5$

22. $1 - b; 2 - a; 3 - d; 4 - c$

23. $P_B = P_E$

$1.3 \times g \times 10 = (0.8 \times g \times X) + (13.6 \times g \times (10 - X))$

$X = 9.6\ cm$

24. $P_B = \rho gh + P_0$

$$P_B = 0.013 \times 10^5 + 10^5$$

$P_B = 1.013 \times 10^5\ N/m^2$

25. $P_C = P_B + \rho g\ h = 1.013 \times 10^5 + 13600 \times 10 \times 10 \times 10^{-2}$

$P_c = 1.149 \times 10^5\ N/m^2$

WORKSHEET – 3

Level – I

1) A	2) C	3) C	4) A	5) C	6) B
7) B	8) A	9) A	10) A		

Level – II

1) A	2) C	3) D	4) A	5) A	6) D	7) B
8) A	9) B	10) D	11) C	12) A	13) B	14) A
15) D	16) A	17) C	18) a-3; b-5; c-1; d-2		19) 2,4	
20) a-3; b-4; c-1,5; d-2		21) 3	22) a-p; b-r; c-q; d-s		23) 2	24) 2
25) 1						

1. $P = \rho gh$
   $= 1000 \times 10 \times 4 = 40000\ \text{N/m}^2$ [option - A]
2. $\frac{\rho_s}{\rho_l} = \frac{x}{x-y}$; $\frac{\rho_s}{\rho_w} = \frac{x}{x-z}$; $\frac{\rho_1}{\rho_w} = \frac{x-y}{x-z}$
   [option - 3]
3. [option - 4]
4. [option - 1]
5. [option - 1]
6. $R \cdot D = \frac{15}{15-12} = 5$ [option - D]
7. $F_b = V_{\text{in}}$ p.g
   $\text{F} = 10 \times 0.5 \times 1000 = 5000\ \text{N}$ [option - B]
8. Liquid Exerts an up thrust on the body. Body will also exert same force on liquid.
   $\therefore \text{p} = \text{p}_0 + \rho gh$ [option -A]
9. $\frac{\rho_s}{\rho_w} = \frac{25}{25-20} = 5$
   [option - B]
10. $\frac{\rho_s}{\rho_l} = \frac{25}{25-13} = \frac{25}{12}$
    $\frac{\rho_l}{\rho_w} = \frac{5}{\left(\frac{25}{12}\right)} = \frac{12}{5}$ [option - D]
11. let m be the mass of copper, in the Ornament then mass of gold (50 m)
    volume of copper $== \frac{\text{m}}{10} = \text{v}_1$
    volume of gold $= \frac{50-m}{20} = v_2$
    loss in weight = Buoyant force
    $(50-46)g = (v_1 + v_2)\rho_w g$
    $4\ \text{g} = \left(\frac{\text{m}}{10} + \left(\frac{50-\text{m}}{20}\right)\right) \times 1 \times \text{g}$
    $4\ \text{g} = \left(\frac{2\ \text{m} + 50 - \text{m}}{20}\right) \times \text{g}$
    $4 = \left(\frac{m+50}{20}\right)$
    $80 = m + 50$
    $\text{m} = 30\ \text{g}$
    [option - C]
12. $F_b = V_{\text{in}}$ p.g $= 10^{-6} \times 1000 \times 9.8$
    $= 9.8 \times 10^{-3}\ \text{N}$
    [option - A]

13. Buoyant force = $(v_{\text{in}} \times \rho_w \times g)$
$= 2.4 \times 10^{-4} \times 1000 \times 10 = 2.4$ N
(OPTION - B)

14. $mg = (v_{\text{in}} \times \rho_{\text{w}} \times g) + (v_{\text{in}} \times \rho_{\text{m}} \times g)$
$10^3 \times 8.6 = 100(10 - X) \times 1 + (100X)(13.6)$
$8600 = 1000 - 100X + 1360X$
$7600 = 1260X$
$X = \dfrac{7600}{1260} \simeq 6$

15. $\text{v}_{\text{in}} = \left(\frac{2}{5} \times 50\right) \times 10 = 200\ \text{cm}^3$
$\text{mg} = \text{v}_{\text{in}} \times \rho_{\text{w}} \times \text{g}$
$\text{m} = 200 \times 1$
$\text{m} = 200$
$\text{density} = \frac{\text{mass}}{\text{volume}} = \frac{200}{500} = 0.4$

16. $w = mg$
[option - D]
$= 200 \times g = 200\ \text{g} \cdot \text{f}$
[option - A]

17. Extra force $= \left(\frac{3}{5} \times 500\right) \times 1 \times g = 300\text{gf}$
[option - C]

18. a-34, b-5, c-1, d-2

19. volume of cylinder $= 240 \times 10^{-6}\ \text{m}^3$
weight in air $= 240 \times 10^{-6} \times 1500 \times 10$
$= 36 \times 10^{-1} = 3.6$ N
[option - B,D]

20. $a - 3; b - 4; c - 1{,}5; d - 2$:

21. [option - 3]

22. p-a; q-c; r-b; s-d:

23. weight of the block = u_1 thrust
$$\left(\frac{A}{5}\right) L \cdot D \cdot g = \left(\frac{3L}{4} \times \frac{A}{5}\right) dg + \left(\frac{L}{4} \times \frac{A}{5}\right) \times 2d \times g$$
$$D = \frac{3d}{4} + \frac{2d}{4} = \frac{5d}{4}$$
[option - B]

24. Total force = weight of liquid +Buoyant force
$= \left(\text{A} \cdot \frac{\text{H}}{2} \cdot \text{d} \cdot \text{g} + \text{A} \cdot \frac{\text{H}}{2} \cdot 2\ \text{d} \cdot \text{g}\right) + \left(\frac{3\text{AL}}{20} + \frac{2\text{AL}}{20}\right) \text{dg}$
$= \frac{3}{2}\text{AHdg} + \frac{1}{4}\text{ALdg}$
$\text{pressure} = \left(\frac{3\text{H}}{2} + \frac{\text{L}}{4}\right) \text{dg} + \text{p}_0 = \left(\frac{6\text{H}+\text{L}}{4}\right) \text{dg} + \text{p}_\circ$
[option - B]

25. Top surface is at depth $\left(\frac{\text{H}}{2} - \frac{3\ \text{L}}{4}\right)$
Force = pressure × Area
$= \text{d} \times \text{g} \times \left(\frac{\text{H}}{2} - \frac{3\ \text{L}}{4}\right) \times \frac{\text{A}}{5}$
$= \left(\frac{\text{H}}{2} - \frac{3\ \text{L}}{4}\right) \text{dg} \frac{\text{A}}{5}$
[option - A]

WORKSHEET – 4

Level – I

1) C	2) C	3) D	4) A	5) C	6) A	7) C
8) B						

Level – II

1) D	2) D	3) C	4) B	5) A	6) B	7) A
8) B	9) C	10) A	11) A	12) A	13) C	14) D
15) C	16) A	17) B	18) A,B,C	19) B	20) A	21) B
22) D	23) 21	24) 23	25) a-q; b-r; c-s; d-p			

1. $V = \frac{\rho_s}{\rho_e} \times 100$
   $V = (0.9) \times 100$
   $= 90\%$
2. $V = \frac{570}{1000} \times 100$
   $= 0.57$
3. $V = \frac{350}{700} = 0.5$
4. Conceptual
5. Conceptual
6. density of cube $= \frac{700}{1000}$
   fracion of volume outside
   the water $= \frac{\rho_w - \rho_s}{\rho_w} = \frac{1-0.7}{1} = 0.3$
   volume of cube outside the water $= 0.3 \times 1000 = 300\ cm^3$
7. [option - A]
   weight of the body = Buoyancy force
   $= \frac{4}{3}\pi(R^3 - r^3) \times \rho \times g = \frac{1}{2}\left(\frac{4}{3}\pi R^3\right)\rho_w g$
   $(R^3 - r^3)\rho = \frac{\rho_w R^3}{2}$
   $\rho = \frac{\rho_w}{2} \cdot \frac{R^3}{R^3 - r^3}$
   $\rho = \frac{1000}{2} \cdot \frac{8^3}{8^3 - 6^3}$
   $\rho = 865\ kg/m^3$
8. $\frac{\rho_s}{\rho_l} = \frac{0.917 \times 10^3}{1.02 \times 10^3} = 0.899$
   $\frac{\rho_s}{\rho_l} \times 100 = 89.9$
   [option - B]
9. Tension = weight - Buoyancy force
   $= 7.6 \times 125 - 125 \times 7.2 = 50gf$
   [option - C]
10. Fraction of volume outside the liquid $= \frac{\rho_l - \rho_s}{\rho_l}$
    $= \frac{1.2 - \rho_s}{1.2} = \frac{3}{8}$
    $1.2 - \rho_s = \frac{3.6}{8}$

$\rho_s = 1.2 - \frac{3.6}{8}$

$= 1.2 - 0.45 = 0.75$ g/cm^3 [option - A]

11. weight of sphere = Buoyant force

$\frac{4}{3}\pi(R^3 - r^3)\rho_s g = \frac{4}{3}\pi R^3 \times \rho \times g$

$(R^3 - r^3)9 = R^3$

$r^3 = \frac{8}{9}R^3$

12. Density of thesolid body $= \frac{150}{250}$

Density of thesolid body $= \frac{3}{5} = 0.6$

$\rho_s < \rho_I$

The body will float

13. let v be the volume of the ball

weight of the ball = $v\rho g$

Buoyancy force on ball = vσg

$F_{net} = v\sigma g - v\rho g = vg(\sigma - \rho)$

vρa = vg$(\sigma - \rho)$

$h' = \left(\frac{\sigma}{\rho} - 1\right)$

it's velocity on reach surface of liquid $v^2 = 2.g\left(\frac{\sigma}{\rho} - 1\right) \cdot h$

The height above water surface, the ball rise is ' h '

$$0^2 - 2.g\left(\frac{\sigma}{\rho} - 1\right) \cdot h = -2gh'\ h' = \left(\frac{\sigma}{\rho} - 1\right)$$

[option - C]

14. Let L be the side of the cube

$\frac{V_{in}}{V} = \frac{\rho_s}{\rho_\omega}$

$\frac{L^2\left(L - \frac{1}{2}\right)}{L^3} = \rho_s$

1- $\frac{1}{2L} = 0$

Weight of thewooden cu be +mg = Bu oyancy force

$L^3\rho_s g + mg = L^3\rho_\omega g$

L$^3\rho_s + 50 =$ L^3

From (1)

$L^3\left(1 - \frac{1}{2L}\right) + 50 = L^3$

L = 10 cm

$\rho_s = 1 - \frac{1}{2\ L} = 1 - \frac{1}{2(10)} = 0.95$

[option - D]

15. Weight of the sphere = Buoyancy force

$\frac{4}{3}\pi(R^3 - r^3)\rho_s g = \frac{4}{3}\pi R^3 \times \rho_w \times g$

$(R^3 - r^3)\rho_s = R^3$

$\rho_s = \frac{R^3}{R^3 - r^3}$

$r = \frac{R}{2}\ \rho_s = \frac{8}{7}$

[option - C]

16. Density of aluminium
$= 2.7 gm/cc$
$= 2.7 \times 10^3 \ kg/m^3$
The Buoyancy force is
$F = \left(\frac{2}{2.7 \times 10^3}\right) \times 1000 \times 9.8$
$F = \frac{19.6}{2.7} = 7.26 \ N$
The reading of above scale
$w_1 - w_2$ = Buoyancy force
$w_2 = w_1$ - Buoyancy force
$= 2 \times 9.8 - 7.26$
$= 19.6 - 7.26 = 12.34 \ N$
The reading of below scale
$= (1 + 2)9.8 +$ Buoyancy force
$= 29.4 + 7.26 = 36.66 \ N$ [option - A]

17. Buoyant force on 1 = Buoyant force on 2
$v \times \rho_l \times g = (2v) \times 0.9 \times g$
$\rho_1 = 1.8$ g/cc [option - B]

18. [option - 1,2,3]

19. $w_{app} = 2 \times 10 \times\times (13.6 - 7.2)$
$= 20 \times 6.4 = 128$ kg

20. [option - A]

21. [option - B]

22. $\frac{v_{in}}{v} = \frac{\rho_s}{\rho_w}$
$v_{out} = \frac{3}{5}v; \ v_{in} = \frac{2}{5}v$
$\frac{\frac{2}{5}v}{v} = \frac{\rho_s}{1}; \ \rho_s = 0.4$
[option - 4]

23. $w = w = (50 \times 10)0.4 \times gf = 200gf$
[option - 1]

24. Extra force $= \frac{3}{5} \times (50 \times 10) \times 1gf = 300gf$ (option - C)

WORKSHEET – 5

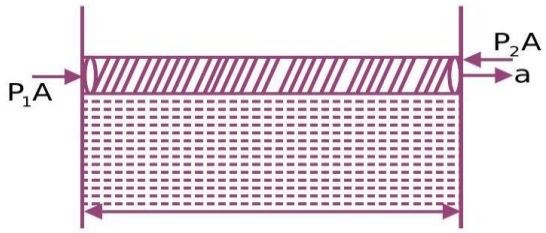
Level – I

1) A	2) A	3) B	4) C	5) A	6) A	7) B
8) B						

Level – II

1) B	2) C	3) B	4) A	5) B	6) A	7) D
8) B	9) B	10) C	11)A	12) B	13) A	14) A
15) C	16) 11	7) D	18) B	19) A	20)B	21) A
22) a-1,3,4; b-2,3; c-2,3; d-5			23) A,B	24) A	25)B	26) C

1. We consider a fluid body of cross-sectional area A (shown in the figure)
$P_1A - P_2A = ma$



or, $(P_1 - p_2)A = \rho Ala$ or,

$P_1 - p_2 = \rho la = 0.92 \times 10 \times 3 = 27.6\text{kPa}$

2. The acceleration of the trolley is $a = g\sin\theta$ down the inclined plane

$a_x = a\cos\alpha = (g\sin\alpha) = g\sin^2\alpha$

Let θ be the angle made by free surface of liquid, then

$\tan\theta = \dfrac{a_x}{g - a_y} = \dfrac{g\sin\alpha\cos\alpha}{g - g\sin^2\alpha}$;

$\theta = \alpha$

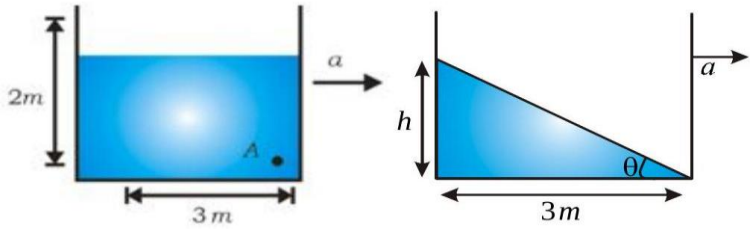
3. Volume of the body $V = \frac{1}{0.8\times10^3} = 1.25 \times 10^{-3}\ \text{m}^3$

mass of thedisplaced water (m) $= 1.25 \times 10^{-3} \times 10^3 = 1.25$ kg

bouyancy force $= \text{m(g} - \text{a)} = 1.25(10 - 1) = 11.25$ N

4. $g_{\text{eff}} = (g + a) = (9.8 + 1.2) = 11\ \text{m/s}^2$

5. $g_{\text{eff}} = (g - a) = (9.8 - 1.2) = 8.6\ \text{m/s}^2$



Volume equality gives

$2 \times 3 = \dfrac{1}{2} \times h \times 3 \Rightarrow h = 4m$

$\therefore \tan\theta = \dfrac{4}{3} = \dfrac{a}{3} \Rightarrow a = \dfrac{4}{3}g$

7. $\rho ghA = \rho LAa, \therefore h = \dfrac{aL}{g}$

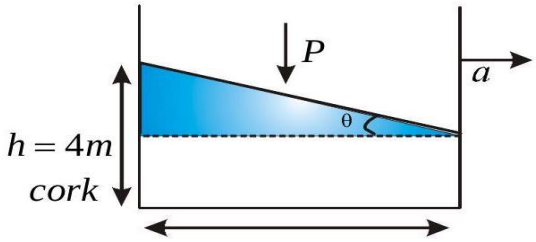
8. When the pressure on cap is 4×10^4 Pa, $h = 4m$

From the figure, $\tan\theta = \frac{3}{\sqrt{3}} \Rightarrow \theta = 60°$

As water does not spill out, the volume of water will be constant.

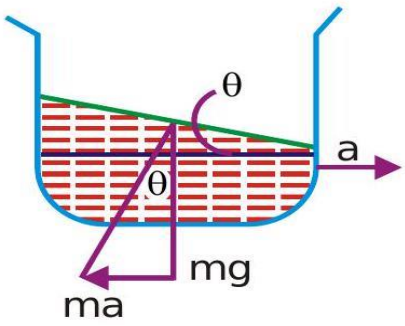
Hence, $2.5 \times \sqrt{3} \times \sqrt{3} = \left(\frac{4+y}{2}\right) \times \sqrt{3} \times \sqrt{3}$

$\Rightarrow y = 1m$



$a = g\tan\theta = 10\sqrt{3}\ \text{m/s}^2$

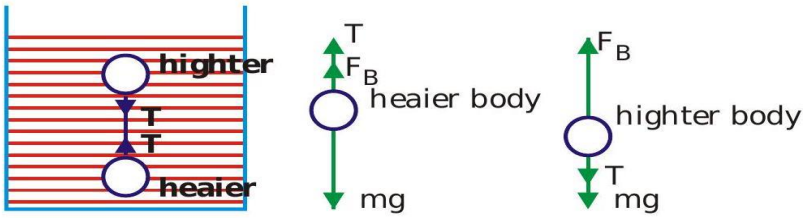
9.



$\tan\theta = \frac{ma}{mg} = \frac{a}{g}$

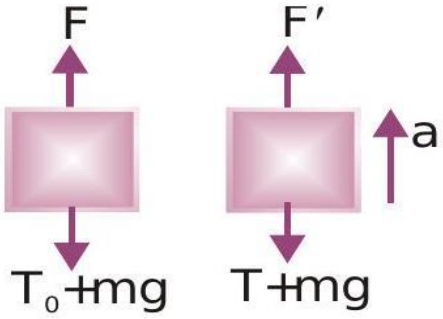
10. Let $m_A = 2m, m_B = 3m$
When block $-B$ is inside the liquid
$a_1 = \frac{m_A g - (m_B g - \text{ upthrust on B})}{m_A + m_B} = \frac{g}{10}$
When block- B is outside the liquid
$a_2 = \frac{m_B g - m_A g}{m_A + m_B} = \frac{3mg - 2mg}{5\ m} = \frac{g}{5}$
Since $a_1 \& a_2$ are constants, motion is periodic, but not simple harmonic

11.



$(250 \times 10^{-6} \times 800\ g) + T = 250 \times 10^{-6} \times f$ liquid $\times g$
$(250 \times 10^{-6} \times 1200g) - T = 250 \times 10^{-6} \times f$ liquid $\times g$
From 1 and 2 we get T = 0.5 N

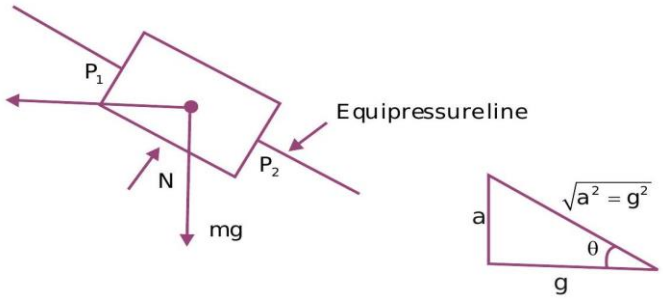
12. Let m be the mass of block. Initially for the equilibrium of block, $F = T_0 + mg$ (1) Hear, F is the upthrust on the block.



When the lift is accelerated upwards, g_{eff} becomes $g + a$ instead of g.
Hence $F' = F\left(\frac{g+a}{g}\right)$
From Newton's second law, $F - T - mg = ma$
Solving Eqs. (1), (2) and (3). We get, $T = T_0\left(1 + \frac{a}{g}\right)$

13. $(40 - 20) = \Delta V \rho_w g \ \therefore \Delta V = \frac{20}{\rho_w \times g} = \frac{20}{10^3 \times 9.8} = 2.04 \times 10^{-3}\ m^3$

14. (i) $F = N = mg\cos\theta$



where $\cos\theta = \frac{g}{\sqrt{g^2+a^2}}$ or $T = \frac{mg^2}{\sqrt{g^2+a^2}}$

(ii) $\tan\theta = \frac{a}{g}$ or $\theta = \tan^{-1}\frac{a}{g}$

15. Let m is the mass of steel piece and v_1 and v_2 are the volume of the cone and steel piece respectively. From condition of floating

$(m + 8.8) \times g = (v_1 + v_2) \times g \times 0.8$

$v_1 = \frac{8.8}{0.5}$; $v_2 = \frac{m}{7.8}$; $(m + 8.8)g = \left(\frac{8.8}{0.5} + \frac{m}{7.8}\right) \times g \times 0.8 \Rightarrow m\left(1 - \frac{0.8}{7.8}\right)$

$= \frac{8.8 \times 0.8}{0.5} - 8.8$

$m\left(\frac{7}{7.8}\right) = 14.08 - 8.8 \Rightarrow \frac{m \times 7}{7.8} = 5.28 \Rightarrow m = 5.88$ kg

16. Therefore, acceleration of the tube = 2a (since COm of the cylinders are moving at ' a ')



$P_A = P_{atm} + \rho(2a)L$ (from the horizontal limb)

$P_A = P_{atm} + \rho gH$ (from the vertical limb)

$\therefore\ a = \frac{gH}{2L}$

17. Acceleration when block B is in the liquid

$$a_1 = \frac{m_A g - (m_B g - \text{upthrust}}{m_A + 3m} = \frac{m_A g - \left(3mg - \frac{3m}{2\rho}\rho g\right.}{m_A + 3m}(\uparrow)$$

Acceleration when B is outside of the liquid

$$a_2 = \frac{3mg - m_A g}{(m_A + 3m)}(\downarrow)\ m_A g - \frac{3}{2}mg = 3mg - m_A g$$

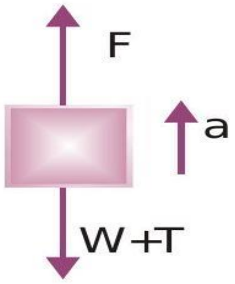
$$\Rightarrow 2m_A g - \frac{9}{2}mg \Rightarrow m_A = \left(\frac{9}{4}\right)m$$

18. F = upthrust force $= V\rho_w(g + a)$

$= \left(\frac{\text{Mass of block}}{\text{density of block}}\right)\rho_w(g + a) = \frac{1}{800}(1000)(11) = 13.75$ N

$F - T - W = ma; 13.75 - 7 - 10 = 1(1);\ T = 2.75$ N

19. Free body diagram of the block is shown in figure.



F = upthrust force $= V_{\rho_\omega}(g+a) = \left(\frac{\text{mass of block}}{\text{density of block}}\right)\rho_\omega(g+a)$

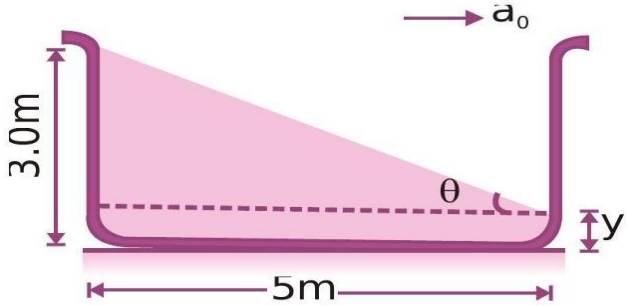
$= \left(\frac{1}{800}\right)(1000)(10+1) = 13.75$ N

20. $W = mg = 10$ N;
Equation of motion of the block is, $F - T - W = ma$
$\therefore 13.75 - T - 10 = 1 \times 1 \therefore T = 2.75$ N

21. When the string is cut $T = 0$
$\therefore a = \frac{F-W}{m} = \frac{13.75-10}{1} = 3.75$ m/s^2

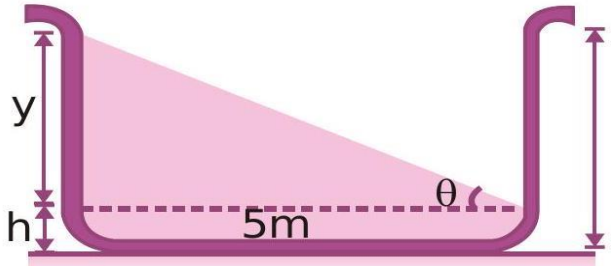
22. $a - 1,3,4; b - 2,3; c - 2,3; d - 5$

23. a) Since the volume of the liquid is constant
Initially volume is final volume $2 \times 5 = \frac{1}{2} \times (3+y) \times 5$



$\therefore y = 1$ m$\tan\theta = \frac{a_0}{g}$ or $\frac{2}{5} = \frac{a_0}{10}$ $\therefore a_0 = 4$ m/s^2

b) If $a_0 = (1.2) \times 4 = 4.8$ m/s^2



$\tan\theta = \frac{a_0}{g}$ or $\frac{y}{5} = \frac{4.8}{10}$ $\therefore y = 2.4$ m

$\therefore h = (3.0 - 2.4) = 0.6$ m

$\therefore$ % of water split over $= \left(\frac{A_1 - A_2}{A_1}\right) \times 100$

$= \left\{\frac{5 \times 2 - (1/2)(3.0+0.6) \times 5}{5 \times 2}\right\} \times 100 = 10\%$

24. $F = PA = [10^5 + (0.3 \times 1030 \times 10)] \times 0.6 \times 0.6 = [10^5 + 3090](0.36) = 3.711 \times 10^4$ N

25. $F_{Up} = PA = [10^5 + (0.3+0.6) \times 1030 \times 10)] \times 0.6 \times 0.6 = 3.933 \times 10^4$ N

26. $T = F_{Up} - F = 0.222 \times 10^4$ N

5. Simple Hormonic Motion

WORKSHEET – 1
Level – I

1. A	4. C	7. D	10. D
2. A	5. B	8. D	11. C
3. B	6. D	9. D	12. D

Level – II

1) A	2) D	3) D	4) A	5) A	6) B	7) C
8) B	9) B	10)C	11)B	12)C	13)C	14)D
15)C	16)D	17) C	18)C	19)B	20)D	21)C
22) A,B	23)a-r; b-p; c-s; d-q		24) A	25) B	26) B	27) A
28) A,B,C	29) a-q; b-s; c-s; d-s					

1. All equations represent S.H.M.
2. In x - y plane all motions are possible.
3. Amplitude is $\sqrt{\mathrm{A}^2 + \mathrm{B}^2}$
4. If phase difference is $\frac{\pi}{2}$, path is circle.
5. Periodic but not simple harmonic.
6. Distance is $A + A + A + A = 4A$.
   Displacement $= 0$.
7. $\frac{dp}{dt} + m\omega^2 y = 0$
   $\frac{d}{dt}(mv) + m\omega^2 y = 0$
   $m\left[\frac{d(v)}{dt} + \omega^2 y\right] = 0$
   $\frac{\mathrm{dv}}{\mathrm{dt}} = -\omega^2 \mathrm{y}$
   $a = -\omega^2 y$
   $\mathrm{a}\alpha = \mathrm{y}$
   Executes simple hormonic motion.
8. $y = \cos^2 wt - \sin^2 wt$
   $y = 1 - \sin^2 wt - \sin^2 wt$
   $y = 1 - 2\sin^2 wt$
   $\frac{dy}{dt} = 0 - 2[2\sin wt.\cos wt.w]$
   $v = -2w[\sin 2wt]$
   $\frac{dv}{dt} = -2w[\cos 2wt.2w]$
   $a = -4\omega^2[\cos 2wt]$
   $a = -4\omega^2[\cos 2wt]$
   $\mathrm{T} = \frac{2\pi}{\omega}$
   $\mathrm{T} = \frac{2\pi}{2\omega}$

$T = \frac{\pi}{\omega}$

9. Amplitude is B.
10. Both restoring and inertia.
11. $W = \frac{\pi}{2}$, $T = 4sec$

Distance covered in 0sec to 3sec is $\rightarrow \ln \frac{3\ T}{4} sec \Rightarrow$ distan ce $= 3a$

12. $x = Asin(\omega t + \pi)$

$x = 0.04sin\left(24\pi t + \frac{\pi}{6}\right)$

13. $x = A\sin \omega t; A = 2$ cm

$x = 2\sin \frac{2\pi}{T} \cdot \frac{T}{6} = \frac{1}{\sqrt{3}}$ cm

14. $x = 0.01\sin 100\pi(t + 0.005)$

$x = 0.01\sin(100\pi t + 0.5\pi)$

$\omega = 100\pi$

$T = \frac{2\pi}{\omega} = \frac{2\pi}{100\pi} = 0.02sec$

15. $x_1 = A_1 \sin wt, x_2 = A_2 \sin(wt + \pi/4)$

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos(\theta)}$$

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos(\pi/4)}$$

$$A = \sqrt{A_1^2 + A_2^2 + \sqrt{2}A_1A_2}$$

16. $y = A\sin \omega t$

in each $\frac{T}{4}$ time interval particle covers distance A.

17. $2\ A = 15 - 5 = 10$ cm

$A = 5$ cm

In time period T sec distance is 4 A.

$$d = 4 \times 5 = 20 \text{ cm}$$

18. $x = A\sin(w.t)$

$T = 4sec, w = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{\pi}{2}$

$x = 4\sin\left(\frac{\pi}{2} \cdot 0.5\right)$

$x = 4\sin\left(\frac{\pi}{4}\right)$

$x = 4 \cdot \frac{1}{\sqrt{2}}$

$x = 2\sqrt{2}$ cm

19. $w = \frac{\pi}{3}$

$T = \frac{2\pi}{3} = \frac{2\pi}{\frac{\pi}{3}}$

$T = 6sec;\ t = 3sec$

$t = \frac{T}{2} sec = \frac{6}{2} = 3sec$

$\Rightarrow$ dis tan ce $= 20$ cm

20. $x = A\sin wt$

$x_1 = A\sin\left(\frac{2\pi}{8}\right) \times 1$

$= \text{Asin}\frac{\pi}{4} = \frac{\text{A}}{\sqrt{2}}$

and $x_2 = A\sin\left(\frac{2\pi}{8}\right) \times 2 = A$

$x_2^1 = x_2 - x_1 = A - \frac{A}{\sqrt{2}} = \frac{(\sqrt{2}-1)a}{\sqrt{2}}$

$\text{Ratio} = \frac{x_1}{x_2} = \frac{\frac{A/\sqrt{2}}{(\sqrt{2}-1)A}}{\sqrt{2}} = \frac{1}{\sqrt{2}-1}$

$= \frac{1}{\sqrt{2}-1} \cdot \frac{\sqrt{2}+1}{\sqrt{2}+1}$

$= \frac{\sqrt{2}+1}{(\sqrt{2})^2-(1)^2} = \frac{\sqrt{2}+1}{2-1} = \sqrt{2}+1$

21. $A = \sqrt{(A_1 - A_3)^2 + (A_2 - A_4)^2}$

$= \sqrt{(8-11)^2 + (6-2)^2}$

$= \sqrt{(4)^2 + (4)^2}$

$A = 4\sqrt{2}$

22. Conceptual

23. Conceptual

24. Particle at rest means $x = 5$

$5 = 5\sin\left(20t + \frac{\pi}{3}\right)$

$t = \frac{\pi}{120}$

25. $a = -\omega^2 x$

$a = -(20)^2 5\sin\left(20t + \frac{\pi}{3}\right)$

$180^0 = \left(20t + \frac{\pi}{3}\right)$

$\pi - \frac{\pi}{3} = 20t$

$t = \frac{\pi}{30}$

26. $\cos\left(20t + \frac{\pi}{3}\right) = -1 \Rightarrow 20t + \frac{\pi}{3} = \pi$

$t = \frac{\pi}{30}$

27. Phase difference between displacement and acceleration is π radius.

28. $v^2 = 180 - 9x^2$

$2\text{v}\frac{\text{dv}}{\text{dt}} = 0 - 9.2\text{x} \cdot \frac{\text{dx}}{\text{dt}}$

$2va = -9.2x \cdot v$

$a = -9x$

$a = -(3)^2 \cdot x$

$\text{w} = 3, \text{T} = \frac{2\pi}{\omega} = \frac{2\pi}{3}$

$a = \omega^2 \cdot x$

$a = (3^2)(3) = 27 \text{ cm/sec}$

29. Conceptual

WORKSHEET – 2
Level – I

1) B	4)A	7)C	10)D
2)B	5)D	8) B	
3)A	6)D	9) A	

1. $0.05 = 0.10\sin(0 + \phi)$
$0.05 = 0.1\sin\phi$
$\frac{1}{2} = \text{Sin}\,\phi$
$\phi = 30°$

Level – II

1)A	2)C	3) C	4) D	5) C	6) A	7)D
8)C	9)C	10)B	11)B	12)A	13)B	14)C
15)B	16)B	17)A	18)B	19)C	20)a-q; b-p; c-s; d-r	
21)1.5	22)B	23)A	24)C	25)A,C	26)A	27)A
28)B	29)D					

2. $\text{V} = \omega\sqrt{\text{a}^2 - \text{y}^2} = 2\sqrt{(60)^2 - (20)^2} = 113$ mm/sec
3. Velocity in mean position $v = a\omega$, velocity at a distance, of half amplitude.

$$V^1 = \omega\sqrt{a^2 - y^2} = \omega\sqrt{a^2 - \frac{a^2}{4}} = \frac{\sqrt{3}}{2}a\omega = \frac{\sqrt{3}}{2}v$$

4. In S.H.M at mean position velocity in maximum so v = aw.
5. $\text{A} = 0.05$ m, $\text{T} = 1.57$sec
$\text{T} = \frac{2\pi}{\omega} \Rightarrow \omega = \frac{2\pi}{\text{T}} = \frac{3.14(2)}{1.57} = 4$ radius /sec
$\text{V}_{\max} = \text{A}\omega$
$= (0.05)(4)$
$= (0.20)$
$= 0.2$ m/s
$a_{\max} = A\omega^2$
$= (0.05)(4)^2$
$= (0.05)(16)$
$a_{\max} = 0.8$ m/s^2
6. $\text{T} = 12$sec
$\Rightarrow \frac{\text{A}}{2} = \text{ASin}\,\omega\text{t} \Rightarrow \frac{\Pi}{6} = \frac{2\Pi}{\text{T}}\text{t}$
$\text{t} = 1$sec
7. $V = \sqrt{144 - 16x^2}$
$V = 4\sqrt{3^2 - x^2}$
Compare $\text{V} = \omega\sqrt{\text{A}^2 - \text{X}^2}$

$a_{\max} = \omega^2 \cdot A$
$a_{\max} = (4)^2(3)$
$= 16 \times 3$
$a_{\max} = 48$ m/s^2

8. $y = 0.2\sin(10\pi + 1.5\pi)\cos(10\pi + 1.5\pi)$
$y = 0.1[2\sin(10\pi t + 1.5\pi)\cos(10\pi t + 1.5\pi)]$
$\mathrm{y} = 0.1\sin 2(10\pi\mathrm{t} + 1.5\pi)$
$\mathrm{y} = 0.1\sin[20\pi\mathrm{t} + 3\pi]$
$\mathrm{W} = 20\pi, \mathrm{T} = \frac{2\pi}{\omega} = \frac{2\pi}{20\pi} = \frac{1}{10} = 0.1\mathrm{sec}$

9. $\mathrm{n} = 12\mathrm{HZ} \Rightarrow \theta = 2\pi\mathrm{n} = 24\pi\mathrm{rad/sec}$ at $\mathrm{t} = 0 \Rightarrow \mathrm{x} = 2$ cm
$\mathrm{x} = 4\mathrm{Sin}(24\pi\mathrm{t} + \phi)$
$\phi = \frac{\pi}{6}\mathrm{rad}$

10. $\mathrm{y}_1 = 10\mathrm{Sin}\left(3\pi + \frac{\pi}{4}\right)$
$y_2 = 5[\sin(3\pi t) + \sqrt{3}\cos 3\pi t]$
$A_1 = 10\ A_2 = 5$

11. $V_1 = \omega\sqrt{A^2 - X_1^2}\ V_1 = \omega\sqrt{A^2 - X_2^2}$
$u_1^2 - u_2^2 = \omega^2(x_2^2 - x_1^2)$
$\mathrm{T} = 2\pi\left[\frac{\mathrm{x}_2^2 - \mathrm{x}_1^2}{\mathrm{u}_1^2 - \mathrm{u}_2^2}\right]$

12. $\mathrm{y} = 5\sin(2\pi\mathrm{t})\mathrm{cm}$
$V = \omega\sqrt{A^2 - X^2}$
and $A = 5$ cm, $\omega = 2\pi, x = 3$ cm
$V = 2\pi\sqrt{(5)^2 - (3)^2}$
$a = 4\pi^2(0)$
$\mathrm{V} = 2\pi(4)$
$a = 0$
$\mathrm{V} = 8\pi\mathrm{cm/s}$
$a = \omega^2 \cdot x$
$a = \omega^2 \cdot A\sin\omega t$
$a = (2\pi)^2.5\sin[2\pi(0.5)]$
$a = 4\pi^2.5\sin\left(2\pi \cdot \frac{1}{2}\right)$

13. $x = a\sin(wt + \pi/6)$
$v = \frac{dx}{dt} = aw\cos(wt + \pi/6)$
$\frac{aw}{2} = aw\cos(wt + \pi/6)$
$\frac{1}{2} = \cos(wt + \pi/6)$
$wt = \pi/6$
$\frac{2\pi}{T} \times t = \frac{\pi}{6}$
$t = \frac{T}{12}$

14. $\mathrm{T} = \frac{\mathrm{T}_1\ \mathrm{T}_2}{\sqrt{\mathrm{T}_1^2 + \mathrm{T}_2^2}} = \frac{\frac{3}{5}.\frac{4}{5}}{\sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2}} = \frac{3}{5} \cdot \frac{4}{5} = \frac{12}{25} = 0.48\mathrm{sec}$

15. Conceptual

16. Conceptual

17. $x = 0.02\sin(3.14t + \pi/2)$
Amplitude $= 0.02$ m

18. Time period $\mathrm{T} = \frac{2\pi}{\omega} = \frac{2\pi}{3.14} = \frac{2(3.14)}{(3.14)} = 2\text{sec}$
19. Frequency $f = \frac{w}{2\pi} = \frac{1}{T} = \frac{1}{2} = 0.5$ Hz
20. Conceptual
21. $A = 50 \text{ mm} = 50 \times 10^{-3} \text{ m}$
    $\mathrm{T} = 2\text{sec},$
    $\omega = \frac{2\pi}{\mathrm{T}} = \frac{2\pi}{2} = \pi\text{rad/s}$
    $\mathrm{V_{max}} = \mathrm{A}\omega$
    $\mathrm{V} = 50 \times 10^{-3} \times (3.14)$
    $\mathrm{V} = 1.57 \times 10^{-1}$
22. $x = 5\cos\left[2\pi t + \frac{\pi}{4}\right]$
    $x = 5\cos[\pi/4]$
    $x = \frac{5}{\sqrt{2}} m$
23. $x = 5\cos(2\pi t + \pi/4)$
    $\frac{\mathrm{dx}}{\mathrm{dt}} = -5\sin[2\pi\mathrm{t} + \pi/4] \cdot 2\pi$
    $V = -10(3.14)\sin(\pi/4)$
    $$\mathrm{V} = 31 \cdot 4 \cdot \frac{1}{\sqrt{2}}$$
    $$= \frac{31.4}{1.414}$$
    $$\mathrm{V} = 22.2 \text{ m/s}$$
24. $\mathrm{V} = -31.4\sin[2\pi\mathrm{t} + \pi/4]$
    $$\frac{\mathrm{dv}}{\mathrm{dt}} = -31.4\cos(2\pi\mathrm{t} + \pi/4).2\pi$$
    $$= -(31.4)2(3.14)\cos(\pi/4)$$
    $$= \frac{-197.192}{\sqrt{2}}$$
    $$= \frac{-197.192}{1.414} = 139.456$$
25. $\mathrm{x} = 0.02\sin(3.14\mathrm{t} + \pi/2)$
    $\mathrm{A} = 0.02 \text{ m}, \omega = 3.14 = \pi\text{rad/s}$
    $\mathrm{V_{max}} = \mathrm{A}\omega = (0.02)(3.14) = 0.68 \text{ m/s}$
    $a_{\max} = -A\omega^2 = (0.02)(3.14)^2 = -0.1972 \text{ m/s}$
26. Conceptual
27. Conceptual
28. $V_1^2 = \omega^2(A^2 - y_1^2)$ and $V_2^2 = \omega^2(A^2 - y_2^2)$
    $V_1^2 - V_2^2 = \omega^2(y_2^2 - y_1^2)$
29. $\frac{V_1^2}{V_2^2} = \frac{A^2 - y_1^2}{A^2 - y_2^2}$
    $$A = \sqrt{\frac{V_1^2 y_2^2 - V_2^2 y_1^2}{V_1^2 - V_2^2}}$$

WORKSHEET – 3

Level – I

1)A	3)B	5)C	7) A
9) D	4)B	6) D	8) B

Level – II

1)C	2)A	3)C	4)B	5)C	6)D	7)B
8)C	9)B	10)B	11)A	12)B	13)B	14)C
15)B	16)C	17)A	18)B,C	19)A	20)a-p; b-q; c-r; d-s	
21)C	22)B	23)A	24)A	25)C	26)D	

27)a-r; b-s; c-q; d-p

1. K.E is maximum at mean position (x = 0).
   $K_{max} = \frac{1}{2} m w^2 A^2$
   But $mw^2 = K = 2 \times 10^6$ N/m.
   A -0.01 m
   $\Rightarrow K_{max} = \frac{1}{2} \cdot K \cdot A^2$
   $= \frac{1}{2}(2 \times 10^6)(0.01)^2$
   $= 10^2$
   $K_{max} = 100$ J
2. $TE = K_{max} + P_{min}$
   $160 = 100 + P_{min}$
3. m = 1 kg, A = 1 m, T = πsec.
   $K.E = \frac{1}{2} m w^2 (A^2 - x^2)$
   $w = \frac{2\pi}{T} = \frac{2\pi}{\pi} = 2 \text{rad/s}, x = 0.8$ m
   $K.E = \frac{1}{2}(1)(2)^2(1^2 - (0.8)^2)$
   $= \frac{1}{2} \times 4 \times (1 - 0.64)$
   $= 2(0.36)$
   $K.E = 0.72$ J
4. at $B, PE = 2KE$ $\frac{KE_A}{KE_B} = 2:1$
   $x_2^2 = \frac{2}{3} A^2$
5. $\frac{A_1}{A_2} = \frac{1}{2} = A_2 = 2$ A,
   But $E_1 = E_2$ and $m_1 = m_2$
   $\frac{1}{2} m_1 w_1^2 A_1^2 = \frac{1}{2} m_2 w_2^2 A_2^2$
   $w_1^2 A_1^2 = w_2^2 (2 A_1)^2$

$w_1^2\ A_1^2 = w_2^2 4 \cdot A_1^2$

$\Rightarrow \frac{w_1^2}{w_2^2} = \frac{4}{1}$

$\Rightarrow \frac{w_1}{w_2} = \frac{2}{1} \Rightarrow 2:1$

6. $x = \frac{A}{2}$

$KE = \frac{1}{2}\ m\omega^2(\ A^2 - x^2)$

$\frac{1}{2}mw^2\left(A^2 - \frac{A^2}{4}\right) = \frac{3}{4} \times \frac{1}{2}mw^2A^2$

7. $y = 6\sin\left(100t + \frac{\pi}{4}\right)$

$A = 6m_1w = 100\text{rad/s}, m = 1\text{ kg}$

$K_{max}\ = \frac{1}{2}mw^2\ A^2$

$= \frac{1}{2}(1)(100)^2(6 \times 10^{-2})^2$

$= \frac{1}{2}(1)(10^4)(36) \times 10^{-4}$

$= 18 \times 10^4 \times 10^{-4}$

$K_{max} = 18\text{ J}$

8. $T.E = K_{max}\ + V_{min}$

$150 = \frac{1}{2}KA^2 + V_{min}$

$150 = \frac{1}{2}(200)(1)^2 + V_{min}$

$150 = 100 + V_{min}$

$PE_{min}\ = V_{min}\ = 50\text{ J}$

9. $K \cdot E_{max} = \frac{1}{2}mw^2A^2$

$= \frac{1}{2}KA^2$

$= \frac{1}{2} \times 2 \times 10^6 \times (0.01)^2$

$K.E_{max} = 10^6 \times 10^{-4}$

$K.E_{max} = 10^2 \times 100\text{ J}$

10. $K.E = P.E$ at $x = \frac{A}{\sqrt{2}}$

Total distance $= \left(\frac{2\ A}{\sqrt{2}}\right)$

$= \sqrt{2} \cdot A$

Time $== \frac{T}{8} + \frac{T}{8} = \frac{T}{4}$

Averege Speed $= \frac{\text{Total dis tance}}{\text{Total time}}$

$= \frac{\sqrt{2} \cdot A}{\frac{T}{4}}$

$= \frac{4\sqrt{2}A}{T}$

11. $A = 10\text{ cm} = 0.1\text{ m}.$

$f = 2H_z, w = 2\pi f$

$= 2\pi(2) = 4\pi f$

$K.E = P.E$ is at $x = A/\sqrt{2}$

$\frac{A}{\sqrt{2}} = A\sin(wt)$

$\sin wt = \frac{1}{\sqrt{2}}$

Phase difference $= \frac{\pi}{4}$

$\Rightarrow \mathrm{x} = 0.1\sin\left(4\pi\mathrm{t} + \frac{\pi}{4}\right)$

12. $\mathrm{P.E} = \frac{1}{2}mw^2x^2$

$x^2 = \frac{2E_1}{mw^2}$

$x = \sqrt{\frac{2E_1}{mw^2}}$

$E_{(x+y)} = \frac{1}{2}mw^2(x+y)^2$

$(x+y)^2 = \frac{2E}{mw^2}$

$(x+y) = \sqrt{\frac{2E}{mw^2}}$

$x + y = \sqrt{\frac{2E}{mw^2}}$

$\sqrt{\frac{2E}{mw^2}} + \sqrt{\frac{2E_2}{mw^2}} = \sqrt{\frac{2E}{mw^2}}$

$\sqrt{E} = \sqrt{E_1} + \sqrt{E_2}$

$E_x = \frac{1}{2}mw^2x^2 = E_1, E_x = \frac{1}{2}mw^2y^2 = E_2$

$\mathrm{y}^2 = \frac{2\mathrm{E}_2}{\mathrm{mw}}$

$y = \sqrt{\frac{2E_2}{mw^2}}$

13. $x = A\sin wt$

$\mathrm{V}_{\max} = \mathrm{Aw}$

$\mathrm{W} = \frac{2\pi}{\mathrm{T}} = \frac{2\pi}{\pi} \times 2 = 4.\,\mathrm{rad/s}$

$\mathrm{V}_{\max} = \mathrm{Aw}$

$1 = \mathrm{A} \times 4$

$\mathrm{A} = \frac{1}{4}$

$\mathrm{F} = \mathrm{ma} = -\mathrm{mw}^2\mathrm{x}$

$= -mw^2A\sin wt$

$= -mw^2A\sin 30°$

$\mathrm{F} = -1\ \mathrm{N}.$

$|F| = 1N.$

14. $A = 1\ \mathrm{m},\ \mathrm{K} = 200\ \mathrm{N/M},\ \mathrm{E} = 150\ \mathrm{J}$

$\mathrm{E} = \mathrm{K}_{\max} + \mathrm{P}_{\min}$

$150 = \frac{1}{2}KA^2 + P \cdot E_{\min}$

$\mathrm{P}_{\min} = 50\ \mathrm{J}$

15. $\mathrm{x} = \mathrm{A}\sin \mathrm{wt}, \mathrm{T} = \sec$

$V = Aw\cos wt$

$\mathrm{K.E} = \frac{1}{2}\mathrm{mv}^2$

$= \frac{1}{2} mA^2 w^2 \cos^2 wt$

Total energy $= \frac{1}{2} \mathrm{mw}^2\ \mathrm{A}^2$

K.E = 75% of total energy

$\frac{1}{2} mw^2 A^2 \cos^2 wt = \frac{75}{100} \times \frac{1}{2} mw^2 A^2$

$\mathrm{Cos}^2\, wt = \frac{3}{4}$

$\mathrm{wt} = \frac{\pi}{6}$

$t = \frac{2\pi}{w} \Rightarrow w = \frac{2\pi}{T}$

$w = \frac{2\pi}{2}$

$\mathrm{t} = \frac{\pi}{6.\pi} = \frac{1}{6}$

16. $\frac{1}{2} K(A^2 - x^2) = \frac{1}{4}\left(\frac{1}{2} KA^2\right)$

$x = \frac{\sqrt{3}}{2} \cdot A$

$\therefore \mathrm{CD} = \frac{\sqrt{3}}{2}\ \mathrm{A}$

$\mathrm{BD} = 2(\mathrm{CD})$

$2(C.B) = \sqrt{3}A.$

17. Conceptual

18. Conceptual

19. Conceptual

20. $\mathrm{E} = \mathrm{K}_{\max} + \mathrm{U}_{\min}$

(c) $20 = K_{\max} + 4$

$\mathrm{K}_{\max} = 16\ \mathrm{J}$

(a) P.Eat $A/2 = 4 + \frac{1}{2} mw^2 x^2$

$= 4 + \frac{1}{2} \mathrm{mw}^2 \left(\frac{\mathrm{A}}{2}\right)^2$

$= 4 + \frac{16}{4}$

$= 4 + 4 = 8\ \mathrm{J}$

(b) $K.E = \frac{1}{2} mw^2 (A^2 - x^2)$

$= \frac{1}{2} mw^2 \left(A^2 - \left(\frac{A}{4}\right)^2\right)$

$= \frac{1}{2} mw^2 \frac{15\ \mathrm{A}^2}{16}$

$= \frac{15}{16}\left(\frac{1}{2} mw^2 A^2\right)$

$= \frac{15}{16}(16) = 15\ \mathrm{J}$

(d) $\mathrm{K.E} = \frac{1}{2} mw^2 (A^2 - x^2)$

$= \frac{1}{2} mw^2 (A^2 - x^2)$

$= \frac{3}{4}\frac{1}{2} mw^2 A^2 K \cdot E = \frac{3}{4}(16^4) = 12J.$

21. $x = 3\sin(0.2\mathrm{t})$

$\mathrm{m} = 3\ \mathrm{g} = 3 \times 10^{-3}\ \mathrm{kg}$

from eq"n equation $w = 0.2\text{red/s}, \text{A} = 3$ m

$\text{K.E} = \frac{1}{2}\text{mww}^2(\text{A}^2 - \text{x}^2)$

$= \frac{1}{2}(3 \times 10^{-3})(0.2)^2\left(3^2 - \left(\frac{3}{3}\right)^2\right)$

$= \frac{1}{2}(3 \times 10^{-3})(0.04)$

$= 0.48 \times 10^{-3}$ J

22. $\text{P.E} = \frac{1}{2}mw^2x^2$

$= \frac{1}{2}(3 \times 10^{-3})(0.2)^2\left(\frac{3}{3}\right)^2$

$= \frac{1}{2}3 \times 10^{-3}(0.04)(1)$

$= 0.06 \times 10^{-3}$ J

23. $T.E = K.E + P.E$

$= (0.48 + 0.06) \times 10^{-3}$

$= 0.54 \times 10^{-3}$ J

24. Conceptual

25. Conceptual

26. Conceptual

27. (a) $x = A\sin(wt)$

$x = A\sin\left(\frac{2\pi}{T} \cdot \frac{T}{4}\right)$

$x = A\sin(\pi/2)$

$\text{X} = \text{A}$

(b) $\frac{A}{\sqrt{2}} = A\sin(wt)$

$\sin wt = \frac{1}{\sqrt{2}}$

$\text{wt} = \frac{\pi}{4}$

$\text{K.E} = \text{P.E.}$

(c) $\frac{3}{4}(K.E) = \frac{1}{4}P.E$

$\frac{3}{4}\left(\frac{1}{2}mw^2(A^2 - x^2)\right) = \frac{1}{4} \cdot \frac{1}{2}mw^2x^2$

$\frac{3}{4}(A^2 - x^2) = \frac{1}{4}x^2$

$3A^2 - 3x^2 = x^2$

$\Rightarrow 4x^2 = 3A^2$

$x^2 = \frac{3A^2}{4}$

$x = \frac{\sqrt{3}A}{A2}$

(d) If $t = \frac{T}{12}$

$x = A\sin\left(\frac{2\pi}{T} \cdot \frac{T}{12}\right)$

$x = A\sin(\pi/6)$

$x = \frac{A}{2}$

$P.E = \frac{1}{2}mw^2x^2$

$= \frac{1}{2}mw^2\left(\frac{A}{2}\right)^2$

$= \frac{1}{8}mw^2A^2$

WORKSHEET – 4
Level – I

1)A	2) D	3) C	4)A	5)C	6) B	7) C
8)	9) C	10) D				

Level – II

1) A	2)B	3)C	4)A	5) B	6) C	7) A
8) A	9)D	10)B	11) C	12)A	13) D	14) B
15)D	16)A	17)C	18) A,B	19)-5	20) a-q; b-t; c-p; d-t	
21)A	22)B	23)A	24)C	25) C	26) A	
27) 5 - longer, 6 - shorter			28) A	29) D	30) C	

1. $T = 2\pi\sqrt{\frac{r}{g}}$

$$T = 2(3.14)\sqrt{\frac{50 \times 10^{-2}}{9.8}}$$

$T = 1.42$sec.

2. $f = \frac{1}{T} = \frac{1}{1.41} = 0.7$hz

3. $T = 2\pi\sqrt{\frac{1}{g}}$

of $\ell^1 = 4\ell$

$$T^1 = 2\pi\sqrt{\frac{\ell^1}{g}}$$

$$T^1 = 2\left(2\pi\sqrt{\frac{\ell}{g}}\right)$$

$T^1 = 2\ T.$

4. Inside the mine ' g ' decreases have $T = 2\pi\sqrt{\frac{\ell}{g}}$

$T \rightarrow$ increuses.

5. $g_{eff} = g + a$

$$T = 2\pi\sqrt{\frac{L}{g_{eff}}}$$

6. $g_{eff} = g - a$

$$T = 2\pi\sqrt{\frac{L}{g_{eff}}}$$

7. $g_{eff} = g$

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

8. At higher temperature lenth of pendulum clock increases resulting increasing time period.
9. Due to force offered by the +ve charge on the plate, the time period will decrease.
10. $T = 2\pi\sqrt{\frac{l}{g}}$

$T = 2\pi\sqrt{\frac{0.4}{9.8}} = 1{,}27\ s$

$(\because g_1 = g); g_2 = g\left(1 - \frac{1600}{6400}\right);\ g_2 = g\left(\frac{3}{4}\right)$

$g_2 = g\left(1 - \frac{d}{R}\right)$

$\frac{T_1}{T_2} = \sqrt{\frac{g_2}{g_1}}$

$\frac{1.27}{T_2} = \sqrt{\frac{g(3/4)}{g}}$

$\frac{1.27}{T_2} = \sqrt{\frac{3}{4}}$

$T_2 = 1.47sec$

11. $\left(\frac{\Delta T}{T}\right)\% = \frac{1}{2}\left(\frac{\Delta l}{l}\right)\%$

$= 0.5\%$

12. $\frac{\Delta\ell}{\ell} \times 100 = 21\%$

$\Rightarrow \frac{T_2}{T_1} = \sqrt{\frac{\ell_2}{\ell_1}}$

$\Rightarrow \frac{T_2^2}{T_1^2} = \frac{121}{100}$

$\frac{T_2}{T_1} = \frac{11}{10} \Rightarrow \frac{T_2}{T_1} - 1 = \frac{11-10}{10}$

$\Rightarrow \frac{T_2 - T_1}{T_1} = \left(\frac{1}{10}\right)$

$\frac{\Delta T}{T} \times 100 = \frac{1}{10} \times 100 = 10\%$

13. $g_{eff} = \sqrt{g^2 + g^2\sin^2\theta + 2g^2\sin\theta\cos(90+\theta)} = g\cos\theta$

$T = 2\pi\sqrt{\frac{L}{g_{eff}}}$

$T = 2\pi\sqrt{\frac{L}{g\cos\theta}}$

14. $g_{eff} = g - \frac{F}{m} = g - \frac{Eq}{m}$

$T = 2\pi\sqrt{\frac{L}{g - \frac{Eq}{m}}}$

15. Restoring force

$F = -mg\sin\theta$

$F = -mg\theta$

$-a = -g\theta$

$= -g\left(\frac{x}{R-r}\right)$

$\omega^2 x = g\left(\frac{x}{R-r}\right);\ \omega^2 = \frac{g}{R-r}$

$\omega = \sqrt{\frac{g}{R-r}}$

$T = \frac{2\pi}{\omega}$

$T = 2\pi\sqrt{\frac{R-r}{g}}$

16. $\Delta t = \frac{\Delta T}{T} = \frac{0.04}{2.04} \times 24 \times 60$ min

$= 28.2$ min, slow

17. Conceptual.

18. Conceptual.

19. w^2 for a pendulum is g/l

when θ is half the amplitude of 60°

acceleration $= -w^2\theta \Rightarrow a = \frac{-10}{1} \times \frac{\pi}{6} = -5\ ms^{-2}$

20. Conceptual.

21. $T = 2\pi\sqrt{\frac{L}{g+a}}$

$T = 2\pi\sqrt{\frac{0.1}{9.8+0.2}}$

$= 2(3.14)\sqrt{\frac{0.1}{10}} = 2(3.14)\sqrt{(0.01)}$

$= 6.28(0.1) = 0.628$

$= 6.28 \times 10^{-1}$

22. $T = 2\pi\sqrt{\frac{0.1}{g-a}}$

$T = 2\pi\sqrt{\frac{0.1}{9.8-0.8}}$

$T = 2\pi\sqrt{\frac{0.1}{9}}$

$T = 2\pi\sqrt{\frac{1}{90}}$

23. $T = 2\pi\sqrt{\frac{\ell}{g}}$

$T = 2\pi\sqrt{\frac{0.1}{9.8}}$

$T = 2\pi\sqrt{\frac{1}{98}}$

24. Conceptual

25. Conceptual
26. Conceptual
27. $T \alpha \sqrt{L}$

$$\frac{T_1}{T_2} = \sqrt{\frac{L_1}{L_2}}$$

$$\frac{T_1}{T_2} = \sqrt{\frac{1.44}{1}}$$

$$\frac{T_1}{T_2} = \frac{1.2}{1}$$

$$\frac{T_1}{T_2} = \frac{12}{10}$$

$T_1 = T_2$

5-longer

6 -shorter

28. $T = 2\pi\sqrt{\frac{L}{g}} = 2\pi\sqrt{\frac{9L}{g}} = 3T$
29. $T' = 2\pi\sqrt{\frac{4\ell}{g}}$

$$T' = 2\left(2\pi\sqrt{\frac{\ell}{g}}\right)$$

$$T' = 2T$$

30. $T' = 2\pi\sqrt{\frac{16\ell}{g}}$

$$T' = 4\left(2\pi\sqrt{\frac{\ell}{g}}\right)$$

$T' = 4T$

WORKSHEET – 5

Level – I

1)A	2)A	3)B	4)D	5)B	6)C	7)D
8)C						

Level – II

1)D	2)C	3)B	4)B	5)A	6)C	7)C
8)C	9)A	10)B	11)C	12)C	13)A	14)A
15)B	16)D	17)D	18)C	19)A	20)C	21)C
22)B	23)A,D	24)a-q; b-p; c-r; d-s		25)B	26)A	27)C
28)A	29)C	30)C				

1. $T = 2\pi\sqrt{\frac{m}{K}} = 2\pi\sqrt{\frac{0.2}{50}} = 0.41\text{sec}$
2. $a_{max} = w^2 \cdot A \cdot T = 3.14\ s$

$= \frac{2\pi}{T} \cdot A\ A = 0.1\ m$

$m = 5\ kg$

$a_{max} = \frac{2\pi}{3.140}(0.1)$

$= \frac{2(3.14)}{(3.14)} \cdot (0.1)$

$a_{max} = 6.2\ m/s^2$

$F_{max} = ma_{max}$

$= 5(0.2) = 1.0$

$F_{max} = 1\ N$

3. $mgx = \frac{1}{2}Kx^2$

$(mg + F)x^1 = \frac{1}{2}Kxx^{1^2}$

$mg = \frac{1}{2}Kx$

$mg + F = \frac{1}{2}Kx$

$A = x^1 - x$

$= \frac{2(mg + F)}{K} - \frac{2mg}{K}$

$= \frac{2mg + 2F - 2mg}{K}$

$A = \frac{2F}{K}$

4. From conservation of energy $mgx = \frac{1}{2}Kx^2$

The maximum displacement of the spring in subsequent motion will be $\frac{2mg}{K}$

From $F = Kx, mg = Kx$ The equilibrium position of the system will occur at the extension of $\frac{mg}{K}$

$\therefore$ amplitude $= \frac{2mg}{K} - \frac{mg}{K} = \frac{mg}{k}$

5. $F = K\Delta x = mg$

$\Delta x = \frac{mg}{K}$

$\Delta x_1 = \frac{(M + m)g}{K}, \Delta x_2 = \frac{mg}{K}$

$\Delta x_1 - \Delta x_2 =$ Amplitude

$\frac{(M + mg)}{K}g - \frac{mg}{K} = A$

$A = \frac{mg}{K}$

6. $T\alpha\sqrt{m}$

$\frac{T_1}{T_2} = \sqrt{\frac{m_1}{m_2}}$

$\frac{1}{0.5} = \sqrt{\frac{200}{m_2}}$

$\frac{1}{0.25} = \frac{200}{m_2}$

$m_2 = 200(0.25)$

$m_2 = 50gm \Rightarrow \Delta m = 200 - 50 = 150gm.$

7. Springs are connected parallel

$K = K_1 + K_2$

$T = 2\pi\sqrt{\frac{m}{K}}$

$f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{K_1 + K_2}{m}}$

$f = \frac{1}{2\pi}\sqrt{\frac{K_1 + K_2}{m}}$

8. $T = 2\pi\sqrt{\frac{M}{K}}$,

$\frac{5T}{3} = 2\pi\sqrt{\frac{M+m}{K}}$

$\frac{5\ T/3}{T} = \frac{2\pi\sqrt{\frac{M+m}{K}}}{2\pi\sqrt{\frac{M}{K}}}$

$\frac{5}{3} = \sqrt{\frac{M+m}{M}}$

$\frac{25}{9} = 1 + \frac{m}{M}$

$\frac{m}{M} = \frac{25}{9} - 1$

$\frac{m}{M} = \frac{16}{9}$

9. When spring of spring constant K is cut into three equal parts each part spring (or) force constant becomes 3 K

$\frac{1}{K} = \frac{1}{3\ K} + \frac{1}{3\ K} + \frac{1}{3\ K}$

10. $\omega = \frac{1}{22}K(x_2^2 - x_1^2)$

$\omega = \frac{1}{2}(5 \times 10^3)(10^2 - 5^2) \times 10^{-4}$

$= \frac{1}{2} \times 5 \times 10^3(75) \times 10^{-4}$

$= \frac{375}{2} \times 10^3 \times 10^{-4}$

$= 184.5 \times 10^3 \times 10^{-4}$

$= 184.5 \times 10^{-1}$

$\omega = 18.45$ J

11. $T = 2\pi\sqrt{\frac{m}{K}}$

$4 = 2\pi\sqrt{\frac{m}{K}}$

$2 = \pi\sqrt{\frac{2}{K}}$

$4 = \pi^2\left(\frac{2}{K}\right)$

$K = \frac{\pi^2}{2}$

$mg = kx$

$2(9.8) = \frac{\pi^2}{2}x$

$\frac{4(9.8)}{\pi^2} = x$

$U = \frac{1}{2}Kx^2 = \frac{1}{2}\frac{\pi^2}{2}\left(\frac{4(9.8)}{\pi^2}\right)^2$

$U \approx 40J$

12. $K_{eff} = K$

$T = 2\pi\sqrt{\frac{m}{K_{eff}}}$

$K_P = K + K = 2\ K$

$\frac{1}{K_{eff}} = \frac{1}{2\ K} + \frac{1}{2\ K} = \frac{2}{2\ K} = \frac{1}{K}$

$K_{eff} = K$

$T = 2\pi\sqrt{\frac{M}{K}}$

13. $(M + m)v = mv_0$

$v = \frac{mv_0}{(M+m)}$

But $v = A\omega$

$\frac{mv_0}{(m+M)} = A\sqrt{\frac{K}{M+m}}$

$A = \frac{mv_0}{(m+M)}\sqrt{\frac{M+m}{K}}$

$A = \frac{mv_0}{\sqrt{K(M+m)}}$

14. $U = \frac{1}{2}\ Kx^2$

$Kx = mg \Rightarrow x = \frac{mg}{k}$

$U = \frac{1}{2}\ K\left(\frac{m^2\ g^2}{k^2}\right)$

$U = \frac{m^2\ g^2}{2k}$

$U_1 = \frac{M^2\ g^2}{2\ K_1}, U_2 = \frac{M^2\ g^2}{2\ K_2}, U_3 = \frac{M^2\ g^2}{2\ K_3},$

15. $T = 2\pi\sqrt{\frac{M}{K}}$

But $\frac{1}{K} = \frac{1}{K_1} + \frac{1}{K_2} + \frac{1}{K_3}$

$T = 2\pi\sqrt{\frac{M}{\left(\frac{1}{K_1} + \frac{1}{K_2} + \frac{1}{K_3}\right)}}$

16. $(K_1 + K_2 + K_3)x = F$

$X = \frac{F}{(K_1 + K_2 + K_3)}$

17. Conceptual

18. Let the compression in spring ' c ' be ' X ' then the extension in each spring A or

$B = \frac{x}{\sqrt{2}} = y$

$F_{\text{net}} = -(Kx + 2Ky\cos 45°)$

$\mathrm{F_{net}} = -(\mathrm{Kx} + \mathrm{Kx}) \left(\because \mathrm{y} = \frac{\mathrm{x}}{\sqrt{2}}\right)$

$\mathrm{F_{net}} = -(2\mathrm{k})\mathrm{x}$

$\mathrm{K_{eff}} = 2\ \mathrm{K}$

$\mathrm{T} = 2\pi\sqrt{\frac{\mathrm{m}}{\mathrm{K_{eff}}}} = 2\pi\sqrt{\frac{\mathrm{m}}{2\ \mathrm{K}}}$

19. $\mathrm{T} = 2\pi\sqrt{\frac{\mathrm{m}}{\mathrm{K}}}$

$\mathrm{m} = \frac{\mathrm{m_1}\ \mathrm{m_2}}{\mathrm{m_1} + \mathrm{m_2}}$

$\mathrm{T} = 2\pi\sqrt{\frac{\mathrm{m_1}\ \mathrm{m_2}}{\mathrm{K}(\ \mathrm{m_1} + \mathrm{m_2})}}$

20. $\frac{1}{2}m_B v_B^2 + \frac{1}{2}m_A v_A^2 + \frac{1}{2}K(2x)^2 =$ energy change

$\frac{1}{2}mv^2 + \frac{1}{2}m(2v)^2 + \frac{1}{2}K4x^2 =$ constan t

differentiating w.r.t to time ' t '

$m(2v)\frac{dv}{dt} + 4m(2v)\frac{dv}{dt} + 4K2x\frac{dx}{dt} = 0$

$m2va + 4m2va = -4k(2x)v$

$2mav + 8mav = -8Kxv$

$10\mathrm{mav} = -8\mathrm{kxv}$

$4\mathrm{ma} = -8\ \mathrm{K}$

$5\mathrm{ma} = -4\mathrm{Kx}$

$a = \frac{-4K}{5m} \cdot A\ (\because x = A)$

$\omega^2 = \frac{4K}{5m};\ \omega = \sqrt{\frac{4K}{5m}}$

$\mathrm{T} = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{5\ \mathrm{m}}{4\ \mathrm{K}}}$

21. Conceptual

22. $\mathrm{f} = \frac{5}{\pi} \Rightarrow \mathrm{T} = \frac{1}{\mathrm{f}} = \frac{\pi}{5}$

$\omega = 2\pi f$

$\omega = 2\pi \cdot \frac{5}{\pi} = 10\mathrm{rad/s} \Rightarrow \mathrm{m} = 20 \times 10^{-3}\ \mathrm{kg}$

$\mathrm{T} = 2\pi\sqrt{\frac{\mathrm{m}}{\mathrm{K}}}$

$\frac{\pi}{5} = 2\pi\sqrt{\frac{20 \times 10^{-3}}{K}}$

$\Rightarrow \frac{1}{10} = \sqrt{\frac{20 \times 10^{-3}}{\mathrm{K}}}$

$$\Rightarrow \frac{1}{100} = \frac{20 \times 10^{-3}}{K}$$
$$= K = 20 \times 10^{-3} \times 100$$
$$K = 2 \text{ N/m}$$

23. $T = 2\pi\sqrt{\frac{m}{K}} = 2\pi\sqrt{\frac{10\times10^{-3}}{4}} = \pi(10^{-1})$
$= 3.14 \times 0.1 = 0.314\text{sec}$
$f = \frac{1}{T} = \frac{1}{0.314} = 3.18\text{sec}^{-1}$

24. Conceptual

25. $K = 1200$ N/m
$m = 3$ kg
$x = 2 \times 10^{-2}$ m
$\text{frequency} = \frac{1}{2\pi}\sqrt{\frac{K}{m}}$
$= \frac{1}{2\pi}\sqrt{\frac{1200}{3}}$
$= \frac{1}{2\pi} \cdot 20 = \frac{10}{\pi} = \frac{10}{3.14} = 3.18 = 3.2 \text{ s}^{-1}$

26. $a = \frac{F}{m} = \frac{Kx}{m} = \frac{1200\times2\times10^{-2}}{3} = 8 \text{ m/s}^2$

27. $V_{\text{man}} = A\omega \Rightarrow \frac{A\omega^2}{\omega} \Rightarrow \frac{a}{\omega} = \frac{8}{2\pi f} = \frac{8}{2\times3.14\times3.2} = 0.4 \text{ ms}^{-1}$

28. Conceptual

29. $U = \frac{1}{2}Kx^2$
$U = \frac{K^2x^2}{2K} = \frac{F^2}{2K}$

30. $E_1 = \frac{1}{2}kx_1^2 \Rightarrow x^2 = \frac{2E_1}{K}$
$E_2 = \frac{1}{2} \text{ kg} \Rightarrow y^2 = \frac{2E_2}{K}$
$E = \frac{1}{2}K(x+y)^2$
$E = \frac{1}{2}K(x^2+y^2+2xy)$
$E = \frac{1}{2}Kx^2 + \frac{1}{2}Ky^2 + \frac{1}{2}2K(x)(y)$
$= E_1 + E_2 + \frac{1}{2} \cdot K \times 2 \times \sqrt{\frac{E_1}{2K}} \cdot \sqrt{\frac{E_2}{2K}}$
$E = E_1 + E_2 + 2\sqrt{E_1E_2}$

6. Optics

WORKSHEET – 1

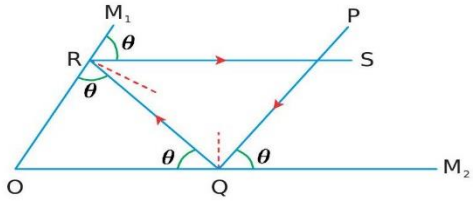
Level – I

1)C,B	4) C	7) B	10) C
2) D	5) A	8) A	
3) C	6) A	9) C	11) A

Level – II

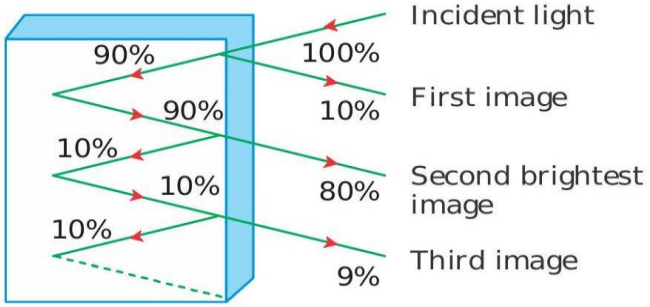
1)B	2)B	3)B	4)B	5)C	6)C	7)C
8)B	9)B	10)C	11)C	12)D	13)A	14)C
15)A	16)B	17)a-1,b-2,c-4,d-3		18)C	19)C	20)D
21)B	22)D	23)C	24)C	25)B	26)A	27) A
28) B	29) B	30) C	31) D			

1. Conceptual
2. When converging beam incident on plane mirror, real image is formed.
3. Relative velocity of image w.r.t man $= 15 - (-15) = 30$ m/s
4. Diminished erect image is formed by convex mirror..
5. If $\frac{360}{\theta}$ = fraction, then n is the integer next higher than
$$\left(\frac{360}{\theta} - 1\right) \quad \therefore n = 7$$
6. Number of images $= \left(\frac{360}{\theta} - 1\right) = \left(\frac{360}{60} - 1\right) = 5$
7. Let the angle between the two mirrors be θ. Ray PQ is parallel to mirror M_1 and R is parallel to M_2.



So $\angle M_1RS = \angle ORQ = \angle M_1OM_2 = \theta$
Similarly $\angle M_2QP = \angle OQR = \angle M_2OM_1 = \theta$
$\therefore$ In $\triangle ORQ, 3\theta = 180^\circ, \theta = 60^\circ$

8. Several images will be formed but second image will be brightest.



9. According to the following ray diagram length of mirror $= \frac{1}{2}(10 + 170) = 90$ cm

10. The walls will act as two mirrors inclined to each other at 90° and so will form $\left(\frac{360}{90}-1\right)=4-1$, i.e., 3 images of the person. Now these images with person will act as object for the ceiling mirror and so ceiling mirror will form 4 images further. Therefore total number of images formed = 3 + 3 + 1 = 7.
    Note: He can see . 6 images of himself.
11. Number of images $=\frac{360^\circ}{\theta}-1$
    Where θ is in degrees, $\therefore 5=\frac{360^\circ}{\theta}=1$
    or $\theta=\frac{360^\circ}{6}=60^\circ$
    New angle, $\theta'=\theta-30^\circ=60^\circ-30^\circ=30^\circ$
    Number of images $=\frac{360^\circ}{30^\circ}-1=11$
12. $\delta=180^\circ-60^\circ=120^\circ$
13. $\angle i=\angle r=0^\circ$
14. When light is reflected from denser medium, a phase difference of a π al ways occurs.
15. When a mirror is rotated by an angle θ, the reflected ray deviates from its original path by angle 2θ.
16. In two images man will see himself using left hand.
17. Conceptual
18. Conceptual
19. It is a case with plane mirror. The image is virtual of the same size and at the same distance at which the object lies.
20. The image always remains at far behind the mirror as the object is in front of him. Suppose the distance of the object from the mirror is x. Then distance of the object from the image will be $2x$. So, when the man runs to wards the plane mirror at vms^{-1}, then the relative speed of image w.r.t him will be 2v i.e., 4 ms^{-1}.
21. There reflected ray turns through angle 2θ, when the mirror is turned through angle θ. Hence rotational speed of the light spot will be 2n revolutions per second.
    $\therefore$ speed of reflected light $=2\pi r\times 2n=4\pi nr$
23. Conceptual
24. $n=\left(\frac{360}{\theta}-1\right)\Rightarrow 3=\left(\frac{360}{\theta}-1\right)\Rightarrow\theta=90^\circ$
25. $f=\frac{R}{2}$ and $R=\infty$ for plane mirror.
26. Subtract the given time from 11:60.
27. $n=\frac{360^\circ}{36}-1=10-1=9$
28. Mirror is rotated at angle 2θ
29. Reflected ray is rotated at angle 2θ so 40°
30. Reflected ray is rotated at angle 2θ so 60°
31. $i=40^\circ\delta=180^\circ-2i=180^\circ-2\times 40^\circ$

WORKSHEET – 2

Level – I

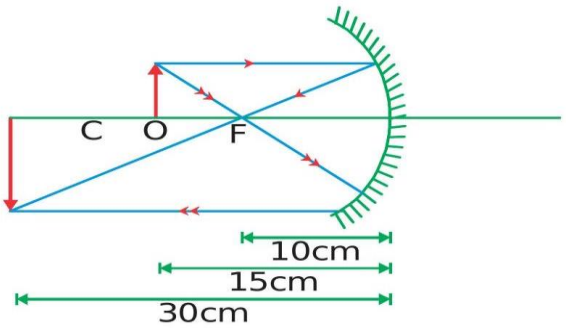
1) D	2) A,C	3) D	4) A	5) D	6) B	7) A
8) B	9) B	10) C	11) B	12) B		

Level – II

1) D	2)B	3) B	4) B	5)4	6)1	7)3
8) D	9)B	10) B	11)C	12)4	13)2	14)2
15)C	16)a-2, b-1,c-3,d-4		17)D	18)1	19)4	20)4
21)A	22)D	23)A	24)C	25)4	26)4	27)2

SOLUTIONS:

1. $f = \frac{R}{2}$
2. $f = \frac{R}{2} = \frac{-0.2}{2} = -0.1$ m
3. $2:1$



$R = 40$ cm; $f = -20$ cm

$u = -30$ cm

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$\frac{1}{-20} = \frac{1}{-30} + \frac{1}{v}$

$\frac{1}{v} = \frac{1}{30} - \frac{1}{20}$

$\frac{1}{v} = \frac{2-3}{60}$

$\frac{1}{v} = \frac{-1}{60}$

$v = -60$ cm

5. $f = \frac{R}{2} \Rightarrow R = 40$ cm
6. Virtual image formed is larger in size in case of concave mirror.
7. Here, object distance, $u = -60$ cm

Magnification, $m = -\frac{v}{u}$

As the image is real

$\therefore m = -\frac{v}{u} = -\frac{1}{2}$ or $v = \frac{u}{2} = -30$ cm

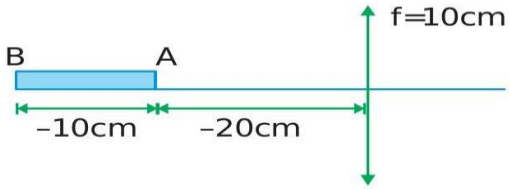
According to mirror formula

$\frac{1}{u} + \frac{1}{v} = \frac{1}{f} \Rightarrow \frac{1}{(-60)} + \frac{1}{(-30)} = \frac{1}{f}$

$\frac{1}{-60} - \frac{1}{30} = \frac{1}{f} \Rightarrow f = -20$ cm

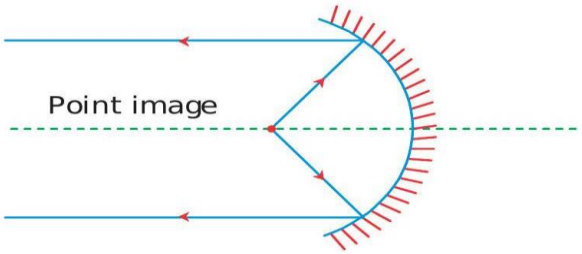
8. When $u_1 = -20$ cm

$\frac{1}{v_1} + \frac{1}{20} = \frac{1}{10}$



$\Rightarrow v_1 = 20$ cm
When $u_2 = -30$ cm
$\frac{1}{v_2} + \frac{1}{30} = \frac{1}{10} \Rightarrow v_2 = 15$ cm
The length of image, $L = V_1 - v_2 = 5$ cm

9. Object should be placed on focus of concave mirror.



10. Concept
11. For $m = 1, u = 2f = 40$ cm
$v = u\left(m = \frac{v}{u}\right)$;
$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$
$\frac{1}{f} = \frac{1}{u} + \frac{1}{u}$
$\frac{1}{f} = \frac{2}{u}$; $u = 2f$
12. Concept
13. $\frac{1}{-15} = \frac{1}{(-2u)} + \frac{1}{u} \Rightarrow u = -7.5$ cm
14. $\frac{l}{O} = \frac{f}{f-u}$; where $u = f + x \therefore \frac{l}{O} = -\frac{f}{x}$
15. Since the image is formed at the point of object i.e. object is kept on the radius of curvature.
$\therefore$ C.F. = F.P. = 60 cm = f
16. m = +ve $\Rightarrow$ virtual image
m = −ve $\Rightarrow$ real image
$|m| > 1 \Rightarrow$ magnified image
$|m| < 1 \Rightarrow$ diminished image
17. Convergence (or power) is independent of medium for mirror.
18. Real, inverted and same in size because object is at the centre of curvature of the mirror.
19. As laws of reflection to be true for all points of the remaining part of the mirror, the image will be that of the whole object. However, as the area of the reflecting surface has been reduced, the intensity of the image iwll reduce (in this case half).
20. $f = \frac{R}{2} = 20$ cm, $m = 2$ For real image; $m = -2$
By Using $m = \frac{f}{f-u,} - 2 = \frac{-20}{-20-u} \Rightarrow u = -30$ cm
For virtual image; $m = +2$
So, $+2 = \frac{-20}{-20-u} \Rightarrow u = -10$ cm

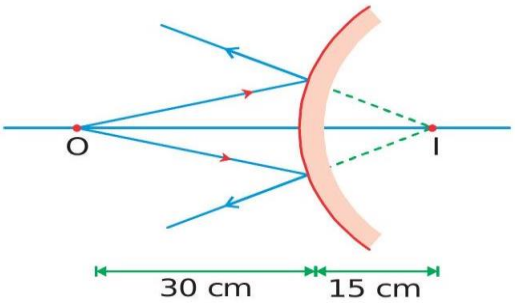
21. According to the mirror formula $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$ or $v=\frac{fu}{u-f}$
When an object is at a distance of 9 m from the concave mirror.
$\therefore u=-9m, f=-1m \Rightarrow v=\frac{(-1)(-9)}{-9+1}=-\frac{9}{8}m$
As the object moves at a constant speed of 5 ms^{-1}, after 1 s the position of image is
$u'=-9m+5m=-4m \Rightarrow v'=\frac{(-1)(-4)}{-4+1}=-\frac{4}{3}m$
The shift in the position of image in 1 s is
$=v-v'=\frac{-9}{8}+\frac{4}{3}=\frac{-27+32}{24}=\frac{5}{24}=\frac{1}{4.8}\approx\frac{1}{5}$
Average speed of the image $=\frac{1}{5}$ ms^{-1}

22. $R=-30 \Rightarrow F=-15$ cm
$O=+2.5$ cm, $u=-10$ cm
By mirror formula $\frac{1}{-15}=\frac{1}{v}+\frac{1}{(-10)} \Rightarrow v=30$ cm
Also $\frac{I}{O}=-\frac{v}{u} \Rightarrow \frac{I}{(+2.5)}=-\frac{30}{(-10)} \Rightarrow I=+7.5$ cm

23. According to Cartesian sign convention
Object distance, $u=-15$ cm
Focal length, $f=-10$ cm
Using mirror formulae $\frac{1}{u}+\frac{1}{v}=\frac{1}{f} \Rightarrow \frac{1}{(-15)}+\frac{1}{v}=\frac{1}{(-10)}$
$\frac{1}{v}=\frac{1}{(-10)}-\frac{1}{(-15)}=\frac{1}{(-10)}+\frac{1}{(15)}$ or $v=-30$ cm
The image is 30 cm from the mirror on the same side of the object.
Magnification, $m=-\frac{v}{a}=-\frac{(-30\text{ cm})}{(-15\text{ cm})}=-2$ cm
The image is magnified, real and inverted.

24. Here focal length $=f$ and $u=-f$
On putting these values in $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$
$\Rightarrow \frac{1}{f}=-\frac{1}{f}+\frac{1}{v} \Rightarrow v=\frac{f}{2}$

25. $u=-30$ cm, $f=+30$ cm, by using mirror formula



$\Rightarrow \frac{1}{+30}=\frac{1}{v}+\frac{1}{(-30)}$
$v=15$ cm, behind the mirror.

26. Image formed by convex mirror is always. Erect diminished and virtual.

27. $m=\frac{f}{f-u} \Rightarrow -3=\frac{f}{f-(-20)} \Rightarrow f=-15$ cm

WORKSHEET – 3

Level – I

1) B	2) C	3) A,B	4) A	5) A	6) A	7) A
8) D	9) B	10) B	11) C	12) D		

Level – II

1. Velocity and wavelength change but frequency remains same.
2. $\mu = \frac{C}{v} = \frac{C}{V\lambda} = \frac{3\times10^8}{4\times10^{14}\times5\times10^{-7}} = 1.5$
3. $\lambda\alpha\frac{1}{\mu} \Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{\mu_2}{\mu_1} = \frac{\mu}{1}$
4. $\text{air } \mu_{\text{water}} = \frac{\text{speed of light in air}}{\text{speed of light in water}} = \frac{c}{v}$
5. $h' = \frac{d_1}{\mu_1} + \frac{d_2}{\mu_2} = d\left(\frac{1}{\mu_1} + \frac{1}{\mu_2}\right)$
6. $v = \frac{\sin i}{\sin r} = \frac{v_1}{v_2}$

   $\Rightarrow \sin r = \sin 30^\circ \times \frac{2u}{u} \Rightarrow \sin r = \frac{1}{2} \times 2 \times 1 \Rightarrow r = 90^\circ$
7. $\text{time} = \frac{\text{distance}}{\text{speed}} = \frac{t}{c/n} = \frac{nt}{c}$
8. Let v' and λ' represent frequency and wavelength of light in medium re spectively.

   $$v' = \frac{u}{\lambda'} = \frac{c/\mu}{\lambda/\mu} = \frac{c}{\lambda} = v$$
9. Apparent depth of bottom

   $= \frac{H/4}{\mu_1} + \frac{H/4}{\mu_2} + \frac{H/4}{\mu_3} + \frac{H/4}{\mu_4}$

   $= \frac{H}{4}\left(\frac{1}{\mu_1} + \frac{1}{\mu_2} + \frac{1}{\mu_3} + \frac{1}{\mu_4}\right)$
10. ${}_a\mu_g = \frac{3}{2}$, ${}_a\mu_w = \frac{4}{3}$

    $\therefore {}_a\mu_g = \frac{{}_a\mu_g}{{}_a\mu_g} = \frac{3/2}{4/3} = \frac{9}{8}$
11. $\text{Apparent depth} = \frac{\text{real depth}}{\mu} = \frac{3}{1.5} = 2 \text{ cm}$

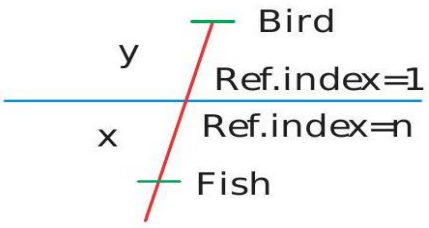
    As image appears to be raised by 1 cm, therefore, microscope must be moved upwards by 1 cm.
12. $\text{The apparent depth ink mark} = \frac{\text{real depth}}{\mu} = \frac{3}{3/2} = 2 \text{ cm}$

    Thus person views mark at a distance $= 2 + 2 = 4$ cm.
13. $\text{Refractive index } \mu = \frac{\text{velocity of light in air}}{\text{Velocity of light in medium}}$

    $\mu_{\text{crown glass}} = 1.5 = \frac{C}{2\times10^8} \Rightarrow C = 1.5 \times 2 \times 10^8 = 3 \times 10^8 \text{ m/s}$

    For dense flint glass $\mu = 1.8 = \frac{3\times10^8}{C_{\text{flint}}}$

    $\therefore C_{\text{flint glass}} = \frac{3\times10^8}{1.8} = 1.67 \times 10^8 \text{ m/s}$
14. The apparent distance of the surface of water for the fish $= x$.



From the surface, the bird is at a distance ny.
∴ The apparent distance = x + ny

15. $\mu = \frac{C}{v} = \frac{3\times10^8}{1.5\times10^8} = 2$
16. Conceptual
17. Refractive index of refracted medium w.r.t. incident medium
$= \frac{\text{speed in incident medium}}{\text{speed in refracted medium}}$
18. When light ray travels from denser to rarer, it deviates away from the normal.
19. Objects are invisible in liquid of R.I. equal to that of object.
20. $\mu\alpha\frac{1}{v} \Rightarrow \frac{\mu_g}{\mu_w} = \frac{v_w}{v_g} \Rightarrow \frac{3/2}{4/3} = \frac{v_w}{2\times10^8}$
$\Rightarrow v_w = 2.25 \times 10^8$ m/s
21. $\lambda_m = \frac{\lambda_a}{\mu} = \frac{C}{V\mu} = \frac{3\times10^8}{5\times10^{14}\times1.5} = 4000$ A
22. $t = \frac{s}{v} = \frac{1.5\times10^8 10^3}{3\times10^8} = 500$ s $= 8.33$ min
23. Apparent rise $= d = \left(1 - \frac{1}{a\mu_w}\right) = 12 \times \left(1 - \frac{3}{4}\right) = 3$ cm
24. Real depth = 1 m
Apparent depth = 1 − 0.1 = 0.9 m
25. $\mu = \frac{h'}{h} \Rightarrow h' = \mu h = \frac{4}{3} \times 18 = 24$ cm
26. $\mu_m = \frac{c}{v} = \frac{n\lambda_a}{n\lambda_m} = \frac{\lambda_a}{\lambda_m}$
27. $t = \frac{\mu x}{c} = \frac{3\times4\times10^{-3}}{3\times10^8} = 4 \times 10^{-11}$ s

WORKSHEET – 4

Level – I

1. B	4. A	7. C
2. C	5. C	8. A
3. D	6. A	9. A

Level – II

1)B	2)D	3)B	4) C	5)A	6)A	7) A
8)C	9)A	10)C	11)C	12)D	13)B	14)C
15)B	16)a-2, b-3,c-4,d-1		17)C	18)C	19)D	20) C
21)B	22)D	23)D	24)D	25) A	26) B	27) A

1. Due to high refractive index its critical angle is very small so that most of the light incident on the diamond is total internally reflected repeatedly and diamond sparkles.

2. ${}_a\mu_g = \frac{1}{\sin C} \Rightarrow \sin C = \frac{1}{{}_a\mu_g}$
   As μ for violet colour is maximum, so $\sin C$ is minimum and hence critical angle C is minimum for violet colour.
3. $\mu = \frac{1}{\sin C} = \frac{1}{\sin 30} = 2$
   $\therefore \text{v} = \frac{3\times10^8}{2} = 1.5 \times 10^8 \text{ m/s}$
4. Ray from setting sun will be refracted at angle equal to critical angle.
5. Optical fibres are used to send signals from one place to another.
6. $\mu_\text{w} < \mu_\text{g} \Rightarrow \text{C}_\text{w} > \text{C}_\text{g}$
7. $\mu = \frac{1}{\sin C} = \frac{1}{\sin 60^\circ} = \frac{2}{\sqrt{3}}$
8. ${}^a_a\mu_\text{g} = \frac{1}{\sin\theta} \Rightarrow \mu = \frac{1}{\sin\theta}$
   Now form Snell's law $\mu = \frac{\sin i}{\sin r} = \frac{\sin\theta}{\sin r}$
   $\Rightarrow \sin r = \frac{\sin\theta}{\mu}$
   From equation (i) and (ii)
   $\sin r = \frac{1}{\mu^2} \Rightarrow r = \sin^{-1}\left(\frac{1}{\mu^2}\right)$
9. $C = \sin^{-1}\left(\frac{1}{\mu}\right)$ and $\mu\alpha\frac{1}{\lambda}$
   Yellow, orange and red have higher wavelengths than green, so μ will be less for these rays, consequently critical angle for these rays will be high,
10. $\text{C} = \sin^{-1}\left(\frac{1}{{}_\text{w}\mu_\text{g}}\right) = \sin^{-1}\left(\frac{\mu_\text{w}}{\mu_\text{g}}\right) = \sin^{-1}\left(\frac{8}{9}\right)$
11. The angle of incidence for B, G and R is 45°. For blue and green, the angle of incidence is greater than the critical angle. Blue and green will be reflected by total internal reflection. Red will be refracted. B and G will not be separated but they are separated from red wihich is refracted.
12. $r = \frac{3h}{\sqrt{7}} = \frac{3\times12}{\sqrt{7}} = \frac{36}{\sqrt{7}}$
13. Here $\sin\text{i} = \frac{1}{\mu} = \frac{3}{5}$ and hence $\tan\text{i} = \frac{3}{4} = \frac{r}{5}$
    This gives = 3 cm, hence diameter = 6 m
14. From the formula $\sin C = \frac{1}{{}_1\mu_2} \Rightarrow \sin C = {}_2\mu_1$
    $= \frac{\mu_1}{\mu_2} = \frac{v_2}{v_1} \Rightarrow \sin C = \frac{10x/t_2}{x/t_1}$
    $= \sin C = \frac{10t_1}{t_2} \Rightarrow C = \sin^{-1}\left(\frac{10t_1}{t_2}\right)$
15. ${}_2\mu_1 = \frac{1}{\sin\theta} \Rightarrow \frac{\mu_1}{\mu_2} = \frac{1}{\sin\theta} \Rightarrow \frac{v_2}{v_1} = \frac{1}{\sin\theta} \Rightarrow \frac{v_2}{v} = \frac{1}{\sin\theta}$
    $\Rightarrow \text{v}_2 = \frac{\text{v}}{\sin\theta}$
16. Conptual
17. Real & apparent depth are explained on the basis of refraction only. TIR not involed here.
18. Semi vertical angle = $C = \sin^{-1}\left(\frac{1}{\mu}\right) = \sin^{-1}\left(\frac{3}{4}\right)$
19. $\mu = \frac{1}{\sin C} \Rightarrow C = \sin^{-1}\left(\frac{1}{2}\right) = 30^\circ$
20. Angle of incidence = 2 (Angle of refraction), $i = 2r$ (given)
    As $\mu = \frac{\sin i}{\sin r} = \frac{\sin 2r}{\sin r}$

$= \frac{2\sin r\cos r}{\sin r}$ [$\because \sin 2\theta = 2\sin\theta\cos\theta$]

or $\cos r = \frac{\mu}{2}$ or $r = \cos^{-1}\left(\frac{\mu}{2}\right)$

$\therefore$ i = 2r = $2\cos^{-1}\left(\frac{\mu}{2}\right)$

21. $\mu = \frac{C}{V} \Rightarrow \mu = \frac{C}{C/2} = 2$ also for total internal reflection

i > C $\Rightarrow$ sin i $\geq$ sin C $\Rightarrow$ sin i $\geq \frac{1}{\mu}$

Hence i $\geq \sin^{-1}\left(\frac{1}{\mu}\right)$ or i $\geq 30°$

22. We know $C = \sin^{-1}\left(\frac{1}{\mu}\right)$

Given critical angle $i_B > i_A$

So $\mu_B < \mu_A$ i.e., B is rarer and A is denser.

Hence light can be totally internally reflected when it passes from A to B.

Now critical angle for A to B

$$C_{AB} = \sin^{-1}\left(\frac{1}{{}_B\mu_A}\right) = \sin^{-1}[{}_A\mu_B]$$

$$= \sin^{-1}\left(\frac{\mu_B}{\mu_A}\right) = \sin^{-1}\left(\frac{\sin i_A}{\sin i_B}\right)$$

23. When total internal reflection just takes place from lateral surface i = C, i.e., 60° = C

$\Rightarrow \sin 60° = \sin C = \frac{1}{\mu} \Rightarrow \mu = \frac{2}{\sqrt{3}}$

Time taken by light to transverse some distance in a medium t = $\frac{\mu X}{c} = \frac{\frac{2}{\sqrt{3}}\times 10^3}{3\times 10^8} = 3.85\mu s$.

24. Here, $v_A = 1.8 \times 10^8$ ms^{-1}

$v_B = 2.4 \times 10^8$ ms^{-1}

Light travels slower in denser medium. Hence medium A is a denser medium and medium B is a rarer medium. Here, light travels from medium A to medium B. Let C be the critical angle between them.

$\therefore \sin C = {}_A\mu_B = \frac{1}{{}_B\mu_A}$

Refractive index of medium B w.r.t medium A is

$${}_A\mu_B = \frac{\text{Velocity of light in mec}}{\text{Velocity of light in me}}$$

$$\therefore \sin C = \frac{v_A}{V_B} = \frac{1.8\times 10^8}{2.4\times 10^8} = \frac{3}{4}$$

$$\Rightarrow C = \sin^{-1}\left(\frac{3}{4}\right)$$

25. $\sin i = \sqrt{\mu_1^2 - \mu_2^2}$

$\sin 30° = \sqrt{(1.5)^2 - \mu_2^2}$

$\frac{1}{2} = \sqrt{(1.5)^2\mu_2^2}$

$\frac{1}{4} = \left(\frac{10}{15}\right)^2 - \mu_2^2$

$\mu_2^2 = \left(\frac{10}{15}\right)^2 \frac{1}{4}$

$\mu_2 = \frac{\sqrt{175}}{30}$

26. $\sin i = \sqrt{\mu_1^2 \mu_2^2}$

$\sqrt{(1.5)^2 - (1.2)^2}$

$\sin i = \sqrt{(2.25(1.44)}$

$\sin i = \sqrt{0.71}$

$i = \sin^{-1}(\sqrt{0.71})$

27

$\sin i \sqrt{\mu_1^2 - \mu_2^2}$

$\sin 6: = \sqrt{\mu_1^2 - (1.2)^2}$

WORKSHEET – 5

Level – I

1) C	4) D	7) C	10) B
2) A	5) A	8) D	
3) B	6) B	9) A	

Level – II

1)C	2)A	3)B	4)A	5)D	6)A	7)B
8)B	9)D	10)D	11)B	12) B	13)C	14)A
15)a-2,b-1,c-4,d-3		16)C	17)B	18)A	19)3	20)D
21)D	22)B	23)C	24)A	25)B		

1. $i = \frac{A+\delta_m}{2} = 50°$
2. In minimum deviation condition $r = \frac{A}{2} = \frac{60}{2} = 30°$
3. In minimum deviation position refracted ray inside the prism is parallel to the base of the prism.
4. For normally emerge $e = 0$
   Therefore $r_2 = 0$ and $r_1 = A$
   Snell's law for incident ray's
   $1 \sin i = \mu \sin r_1 = \mu \sin A$
   For small angle, $i = \mu A$
5. Deviation = zero
   So, $\delta = \delta_1 + \delta_2 = 0$
   $(\mu_1 - 1)A_1 + (\mu_2 - 1)A_2 = 0$
   $A_2(1.75 - 1) = -(1.5 - 1)15°$
   $A_2 = \frac{0.5}{0.75} \times 15^0$
   $A_2 = -10°$
6. Here, $i = 45°, A = 60°$
   Angle of minimum deviation
   $\delta_m = 2i - A = 2 \times 45^0 - 60° = 30°$
7. $\mu = \frac{\sin i}{\sin A/2} \Rightarrow \sqrt{2} = \frac{\sin i}{\sin\left(\frac{60}{2}\right)}$
8. $\tan i = \mu$ (by Snell's law)
9. $\frac{\delta_1}{\delta_2} = \frac{A_1}{A_2}$

10. In minimum deviation i = e = 30°, so angle between emergent ray and sec ond refracting surface is $90° - 30° = 60°$
11. For different colours μ change so deviation of different colour's also different
12. A green leaf reflects green light and so it appears to be green. When incident light has no wave length corresponding to green in appear black.
13. Red colour has more were length, so i devites less.
14. $\mu = \cot\left(\frac{A}{2}\right)$

$$\mu = \frac{\sin\left(\frac{A + D_m}{2}\right)}{\sin\frac{A}{2}}$$

$$\frac{\cos\frac{A}{2}}{\sin\frac{A}{2}} = \frac{\sin\left(\frac{A + D_m}{2}\right)}{\sin\frac{A}{2}}$$

$$\cos\frac{A}{2} = \sin\left(A + \frac{D_m}{2}\right)$$

$$= \sin\left(90 - \frac{A}{2}\right) = \sin\left(\frac{A + D_m}{2}\right)$$

$D_m = 180 - 2A$

15. Conceptual
16. Conceptual
17. $\delta = (\mu - 1)\mathrm{A}$

$5^0 = (1.5 - 1)\mathrm{AA}$

$A = \frac{50}{0.5} = 10°$

18. Angle of refracion is 30° for both red and violet.
19. $\omega = \frac{\delta_v - \delta_r}{\delta_y - 1} = \frac{1.65 - 1.61}{1.63 - 1}$
20. $\mathrm{i} = \mathrm{eA} = 60°$

$i = e = \frac{3}{4}A^0$

$\mathrm{i} = \mathrm{e} = \frac{3}{4}(60) = 45°$

$\mathrm{S_T}$ and $\mathrm{r_T} = -\mathrm{d_m} = \mathrm{d}$

$A + d_m = 2i$

$\mathrm{d_m} = 2\mathrm{i} - \mathrm{A}$

$= 2(45) - 60 = 30°$

21. $\mathrm{A} = 60°, \mathrm{i} = \mathrm{e} = 45°$ By $\mathrm{i} + \mathrm{e} = \mathrm{A} + \delta$, we get

$\Rightarrow 45 + 45 = 60 + \delta \Rightarrow \delta = 30°$

22. $\mathrm{i} = 45°; \mathrm{A} = 60°; \delta_{\mathrm{m}} = 2\mathrm{i} - \mathrm{A} = 30°$
23. In minimum deviation position $\angle \mathrm{i}_1 = \angle \mathrm{i}_2$ and $\angle \mathrm{r}_1 = \angle \mathrm{r}_2$.
24. $\delta\alpha(\mu - 1)$
25. $A = 60°, \delta_m = 30°$ so $\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$

$$\mu = \frac{\sin\left(\frac{60° + 30°}{2}\right)}{\sin\left(\frac{60°}{2}\right)} = \frac{\sin 45°}{\sin 30°} = \sqrt{2}$$

Also $\mu = \frac{1}{\sin C} \Rightarrow C = \sin^{-1}\left(\frac{1}{\mu}\right) \Rightarrow C = 45°$

$\therefore i = 2c = 60°$

Applying S well's law, $1\sin 60 = 2\sin\gamma$

$\sin\gamma = \frac{\sqrt{3}}{4} \Rightarrow \gamma = \sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$

WORKSHEET – 6

Level – I

1. A
2. D
3. A
4. A
5. B
6. D

Level – II

1) A 2) B 3) A 4) D 5) A 6) B 7) A
8) C 9) D 10) B 11) B 12) C 13) D 14) A
15) D 16) D 17) A 18) C 19) 0O
20) a-2;b-1;c-4;d-3 21) B 22) B 23) B 24) B 25) A
26) D 27) C 28) C 29) C 30) C 31) A
32) B

1 to 26 conceptual

27. A convex lens will act as converging when it is immersed in aliquid which has a lesser refractive index than the lens the lens and act as diverging when the sur rounding medium has a greater refractive index than it Here water has a lesser refractive index than lens and carbondisulphide has a highter than it.
28. Conceptual
29. $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

$$\frac{1.52}{v} - \frac{1}{(-8)} = \frac{1.52-1}{2}$$

$$\frac{1.52}{v} + \frac{1}{8} = \frac{0.52}{2}$$

$$\frac{1.52}{v} = \frac{0.52}{2} - \frac{1}{8}$$

$v = 11.3$ cm

30. $m = \frac{\mu_1 v}{\mu_2 u} = \frac{1(11.3)}{(1.52)(8)} = -0.929$
31. $\frac{1.33}{8} + \frac{1.52}{v} = \frac{1.52-1.33}{2}$

$v = -21.3$ cm

32. $m = \frac{n_1 v}{n_2 u} = \frac{(1.33)(-21.3)}{(1.52)(-8)}$

$m = 2.33$

WORKSHEET – 7

Level – I

1) C 2) B 3) A 4) A,B,C,D 5) A 6) A

Level – II

1)A 2)B 3)A 4)A 5)B 6)A 7)A
8)B 9)A 10)A,B,C,D 11)D 12)A 13)A 14)B
15)D 16)a-1,b-2,c-3,d-4 17)B 18)B 19)D 20)-30 cm
21)A 22)A 23)B 24)A 25)B 26)A 27)C

1. $R_1 = R$
2. $R_2 = R$
3. $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$
4. $R_1 = \infty$
5. $R_2 = -R$
6. $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right), R_1 = \infty R_2 = -R$
7. $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right), R_1 = -eR_2 = +R$
8. $\frac{1}{f} = \frac{(\mu-1)}{R}$

$$\frac{1}{f} = \frac{R}{(\mu - 1)} = \frac{20}{2.5 - 1} = \frac{20}{1.5} = 13$$

9. $f = \frac{R}{2(\mu-1)} = \frac{5}{2(2.5-1)} = \frac{5}{2(1.5)} = \frac{5}{3} = 1.6$
10. $f = \frac{R}{\mu-1}$

$$10 = \frac{R}{2.5 - 1}$$

$$10 = \frac{R}{1.5}$$

$R = 10 \times 1.5 = 15$ cm

11. $f = \frac{R}{2(\mu-1)}$

$$5 = \frac{R}{2(2.5 - 1)} = \frac{R}{2(1.5)} = \frac{e}{3}$$

$R = 15$

12. $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)\ R_1 = \infty, R_2 = R$

$$f = -\frac{R}{\mu - 1}$$

13. $m_L = 4$
14. Using lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, for the first lens $u = -4m, f_1 = 2m \therefore v_1 = 4m$. So u for 2 nd lens is equal to +1 m. Again using lens formula for second lens putting $u_2 = 1$ m and $f_2 = 1$ m we get $v_2 = 0.5$. Therefore distance form object $= 7 + 0.5 = 7.5$ m
15. For real image $m = 2$

$$\therefore m = \frac{f}{u + f} \Rightarrow -2 = \frac{f}{u + f} = \frac{20}{u + 20} \Rightarrow u = -30 \text{ cm}$$

16. Conceptual
17. $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$$\Rightarrow \frac{1}{+10} = (1.5 - 1)\left(\frac{1}{+7.5} - \frac{1}{R_2}\right) \Rightarrow R_2 = -15 \text{ cm}$$

18. $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$$= (1.5 - 1)\left(\frac{1}{10} + \frac{1}{10}\right) = \frac{1}{10} \Rightarrow f = 10 \text{ cm}$$

$\therefore$ Radius of curvature of concave mirror $= 2f = 20$ cm

19. $\frac{1}{F} = (1.5 - 1)\left(\frac{1}{20} - \frac{1}{\infty}\right) \Rightarrow F = 40$ cm
20. $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$$\frac{1}{u}=\frac{1}{v}-\frac{1}{f}=\frac{1}{-10}-\frac{1}{-15}=\frac{-3+2}{30}=\frac{-1}{30}=-30 \text{ cm}$$

21. $\frac{1}{\mathrm{f}}=\left({}_{\mathrm{g}}\mu_{\mathrm{a}}-1\right)\left(\frac{1}{\mathrm{R}_1}-\frac{1}{\mathrm{R}_2}\right)=\left(\frac{2}{3}-1\right)\left(\frac{2}{10}\right)$
$=$ F $=-15$ cm, so behaves as concave lens

22. By formula $\frac{1}{\mathrm{f}}=(\mu-1)\left(\frac{1}{\mathrm{R}_1}-\frac{1}{\mathrm{R}_2}\right)$
$$=(1.5-1)\left(\frac{1}{40}+\frac{1}{40}\right)=0.5\times\frac{1}{20}=\frac{1}{40}$$
$\therefore$ f $=40$ cm

23. Applying the lens makers formula
$\frac{1}{f}=P=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$
We know that the greater positive power is for that material for which $\frac{1}{R_1}-\frac{1}{R_2}$ is maximum and positive. For this condition R_1 and R_2 should be small as possible but still it must be positive, therefore we must select the combination which has less radius of curvature for convex lens.
Hence option 2 is correct.

24. Here u = mf
For divergent lens (concave)
$\therefore$ v $=\frac{\mathrm{mf}}{\mathrm{m}+1}$
Now magnification $=\frac{\mathrm{v}}{\mathrm{u}}=\frac{\mathrm{mf}}{\mathrm{m}+1}\times\frac{1}{\mathrm{mf}}=\frac{1}{\mathrm{m}+1}$

25. $\frac{\mathrm{I}_1}{\mathrm{O}}=\frac{\mathrm{v}}{\mathrm{u}}$ and $\frac{\mathrm{I}_2}{\mathrm{O}}=\frac{\mathrm{v}}{\mathrm{u}}\Rightarrow \mathrm{O}^2=\mathrm{I}_1\mathrm{I}_2$

26. $\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$
Given $R_1=+20$ cm, $R_2=-20$ cm, $\mu=1.5$
$=$ f $=20$ cm. Parallel rays converge at focus. so L = f.

27. $\mu=\frac{\lambda_{\mathrm{a}}}{\lambda_{\mathrm{m}}}=\frac{3}{2}$
$\frac{1}{f}=\frac{\mu-1}{R}=\frac{1}{2R}\Rightarrow\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$
$\Rightarrow\frac{1}{8}-\frac{1}{-24}=\frac{1}{2\mathrm{R}}\Rightarrow\frac{3+1}{24}=\frac{1}{2\mathrm{R}}$
$\Rightarrow$ R $=3$ m

WORKSHEET – 8

Level – I

1)A	3)A	5)A
2)B	4)B	6)D

Level – II

1)A	2)A	3)A	4)A	5)A	6)B	7)C
8)D	9)B	10)A	11)B	12)B	13)A	14)A
15)A	16)A	17)A	18)5D	19)D	20)a-r;b-s;c-p;d-q	
21)C	22)A	23)C	24)A	25)D	26)B	

1. $F=80$ cm, $\mathrm{f}_1=40$ cm

Using $\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}$, we get

$$\frac{1}{f_2}=\frac{1}{F}-\frac{1}{f_1}=\frac{1}{80}-\frac{1}{40}=\frac{1-2}{80}=-\frac{1}{80}$$

$\therefore \mathrm{f}_2=-80$ cm

$$\therefore P=\frac{100}{f_2}=\frac{100}{-80}=-1.25\mathrm{D}$$

2. $P_1=+6D, P_2=-4D$

$\therefore P=P_1+P_2=6-4=2D$

$$\therefore \mathrm{f}=\frac{100}{\mathrm{P}}=\frac{100}{2}=+50 \text{ cm}.56$$

3. $f=\frac{100}{P}=\frac{100}{-4.5}=-22.2$ cm; $n=1.6$

Let $R_1=-R$ and $R_2=+R$

$$\therefore \frac{1}{f}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)=(n-1)\left(-\frac{2}{R}\right)$$

$\therefore \mathrm{R}=-2(\mathrm{n}-1)\mathrm{f}=-2(1.6-1)\times 22.2$

$=-2\times 0.6\times 22.2=-26.6$ cm

4. For equiconvex lens

$\mathrm{R}_1=+\mathrm{R}$, $\mathrm{R}_2=-\mathrm{R}$

lens Maker's formula

$$\frac{1}{f}=(\mu-1)\left(\frac{1}{R}+\frac{1}{R}\right)=\frac{2(\mu-1)}{R}$$

For each half lens $R_1=R, R_2=-\infty$

$$\frac{1}{f}=(\mu-1)\left(\frac{1}{R}-\frac{1}{-\infty}\right)=\frac{\mu-1}{R}$$

From (1) and (2) we get $\frac{f'}{f}=2$

$f=2\mathrm{f}=2\times 15=30$ cm

5. Here $\mu=1.5, R_1=+40$ cm $=0.40$ m, $R_2=-40$ cm $=-0.40$ m

$$P=\frac{1}{f}=(\mu-1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$$

$$=(1.5-1)\left[\frac{1}{0.40}+\frac{1}{0.40}\right]=2.5D$$

6. Let $R_1=+R$ and $R_2=-R$, Then

$$P=\frac{1}{f}=(\mu-1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$$

$$\text{or } 5=(1.5-1)\left[\frac{1}{R}+\frac{1}{R}\right]=0.5\times\frac{2}{R}$$

$$\therefore R=\frac{0.5\times 2}{5}\text{ m}=\frac{1}{5}\text{ m}=20\text{ cm}$$

7. Here $R_1=20$ cm $=0.2$ m

$\mathrm{R}_2=-20$ cm $=-0.2$ m, $\mu_\mathrm{g}=1.5, \mu_1=1.25$

For the lens placed in air:

$$P_1=(\mu_g-1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]=(1.5-1)\left[\frac{1}{0.2}-\frac{1}{-0.2}\right]=5\mathrm{D}$$

For the lens placed in liquid:

$$P_2=\left(\frac{\mu_g}{\mu_1}-1\right)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]=\left(\frac{1.5}{1.25}-1\right)\left[\frac{1}{0.2}-\frac{1}{-0.2}\right]=2\mathrm{D}$$

$$\therefore \frac{P_1}{P_2}=\frac{5}{2}=5:2$$

8. $f = +0.5$ m

$$P = \frac{1}{f} = \frac{1}{+0.5} = +2D$$

9. $R_1 = +10$ cm, $R_2 = -15$ cm, f $= +12$ cm

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{12} = (\mu_g - 1)\left(\frac{1}{10} - \frac{1}{15}\right) = (\mu_0 - 1) \times \frac{1}{6}$$

$(\mu - 1) = 0.5$

10. $\mathrm{f_1} = +10$ cm $= 0.10$ m

$\mathrm{f_2} = -5$ cm $= -0.05$ m

$$\frac{1}{\mathrm{f}} = \frac{1}{\mathrm{f_1}} + \frac{1}{\mathrm{f_2}} = \frac{1}{0.10} + \frac{1}{-0.05} = -10$$

$$\mathrm{f} = -\frac{1}{10} \text{ m} = -10 \text{ cm}$$

11. $\mathrm{f_1} = 20$ cm $= 0.20$ m

$\mathrm{f_2} = -50$ cm $= -0.50$ m

$$\frac{1}{\mathrm{f}} = \frac{1}{\mathrm{f_1}} + \frac{1}{\mathrm{f_2}} = \frac{1}{0.20} + \frac{1}{-0.50}$$

$$= 5 - \frac{1}{0.50} = 5 - 2 = 3$$

12. Here $\mathrm{f_1} = +10$ cm $= 0.10$ m,

$\mathrm{f_2} = -5$ cm $= -0.05$ m

$$\frac{1}{\mathrm{f}} = \frac{1}{\mathrm{f_1}} + \frac{1}{\mathrm{f_2}} = \frac{1}{0.10} + \frac{1}{-0.05} = -10$$

$$f = \frac{1}{10} m = -10 \text{ cm}$$

$$P = P_1 + P_2 = \frac{1}{f_1} + \frac{1}{f_2} = -10D$$

13. Here $\mathrm{f_1} = +50$ cm

$f_2 = -50$ cm

As $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$

$$\frac{1}{f_2} = \frac{1}{f} - \frac{1}{f_1} = \frac{1}{-50} - \frac{1}{50} = -\frac{1}{25}$$

$f_2 = -25$ cm $= -0.25$ m

The second lens is a diverging lens. Its power is $P_2 = \frac{1}{f_2} = \frac{1}{-0.25m} = -4D$

14. $P = P_1 + P_2 = +15 - 5 = 10D$

15. $\mathrm{P} = \mathrm{P_1} + \mathrm{P_2} = +15 - 5 = 10\mathrm{D}$

Focal of combination

$$\mathrm{f} = \frac{1}{\mathrm{P}} = \frac{1}{10} \text{ m} = 10 \text{ cm}$$

16. $\mathrm{P} = \mathrm{P_1} + \mathrm{P_2}$

$$= \frac{100}{f_1(\text{ cm})} + \frac{100}{f_2(\text{ cm})}$$

$$= \frac{100}{25} + \frac{100}{25} = 4 - 5 = -1D$$

17. $m = -3, P = 4D, f = \frac{1}{4} m = 25$ cm

$$m = \frac{f}{f+4} - 3 = \frac{25}{25+4}$$

$$u = \frac{-100}{3} = -33.33 \text{ cm}$$

18. $P = P_1 + P_2 = 10 - 5 = 5D$

19. $P = P_1 + P_2 = 10 - 5 = 5D$

$$f = \frac{1}{P} = \frac{1}{5} m = 20 \text{ cm}$$

$$m = \frac{f}{f+u} = \frac{20}{20+u}$$

$$u = -10 \text{ cm}$$

20. Conceptual

21. $P = \frac{100}{f(\text{ in cm})}$

$$f = \frac{100}{10} = 10 \text{ cm}$$

22. $\frac{1}{f_a} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$= (1.5 - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

and $\frac{1}{f_m} = \left(\frac{\mu_g - \mu_m}{\mu_m}\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$= \left(\frac{1.5 - 1.6}{1.6}\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$\frac{f_m}{f_a} = \frac{(1.5-1)}{\frac{-0.8}{1.6}} = -8$

$f_m = -8 f_a = -8 \times \left(\frac{-1}{5}\right) = 1.6m$

$\left[\because f_a = \frac{1}{p} = \frac{-1}{5} \text{ m}\right]$

$\therefore P_m = \frac{\mu}{f_m} = \frac{1.6}{1.6} = 1D$

23. $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$

$P = \frac{f_1 + f_2}{f_1 f_2}$

24. If a biconvex lens is cut perpendicular to the principal axis, then the focal length becomes double.

$f' = 2f$

New power $P' = \frac{1}{2f} = \frac{P}{2}$

$\left[\because \frac{1}{f} = P\right]$

$= \frac{4}{2} = 2D$

25. When both lenses are combined together, resultant focal length is given by

$$\frac{1}{f} = \frac{1}{8} - \frac{1}{5} = -\frac{3}{40}\ \text{cm}^{-1}$$

$$\Rightarrow \frac{1}{f} = -\frac{30}{4}\ \text{m}^{-1}$$

$$\text{or}\ \ P = \frac{1}{f} = -7.5\text{D}$$

(Because focal length of convex is positive and concave is negative)

26. When two thin lenses are in contact their powers get added algebraically. We say that

$$P = P_1 + P_2$$

$$\Rightarrow \frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} \Rightarrow \frac{1}{f} = \frac{f_1 + f_2}{f_1 f_2}\ \text{ or}$$

$$f = \frac{f_1 f_2}{f_1 + f_2}$$